



Blockchain-based real-estate tokenization in regulated markets: A systematic review and socio-technical governance framework

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Abstract

Existing research on blockchain-based real-estate tokenization remains divided among technical architectures, ownership and token models, institutional regulation, and adoption studies, leaving these elements insufficiently integrated for the design of regulated marketplaces. This paper presents a mixed-method study combining a PRISMA-based systematic review of 63 primary studies with an exploratory perception survey of 153 respondents in Saudi Arabia, a highly regulated real-estate market. The review develops a taxonomy of blockchain-based real-estate tokenization systems across eight dimensions: architecture, ownership, transactions, tokenization, certification, stakeholders, technology, and adoption barriers. Findings show movement from registry-centric digitization toward marketplace-oriented models and identify regulated-deployment conditions spanning legal enforceability, identity assurance, governance, privacy, interoperability, continuous security monitoring, and long-term cryptographic resilience. Survey responses show a coherent, predominantly cautious orientation. Reliability and single-factor results identify a general response disposition, while open responses distinguish recognition of particular capabilities from endorsement of their wider adoption consequences. The study contributes an evidence-traceable synthesis that connects the taxonomy, cross-dimensional deployment gaps, exploratory perception evidence, and a socio-technical governance framework comprising a jurisdiction-gated adoption pathway and twelve design requirements, with applicability assessed against documented institutional initiatives.

Keywords Blockchain · Real-estate tokenization · Systematic literature review · Regulatory compliance · Dynamic security monitoring · Socio-technical governance

1 Introduction

Real-estate tokenization sits at the intersection of blockchain, digital-asset markets, and regulated institutional governance. It represents property-related rights, claims, certificates, or investment interests as verifiable and transferable blockchain tokens governed through programmable infrastructure. For a high-value, structurally complex asset class, this model can improve auditability, document integrity, transaction traceability, conditional execution, and digital coordination (Saari et al. 2022a; Avci and Erzurumlu 2023; Banerjee et al. 2022; Yadav and Kushwaha 2021; Gutierrez and Xu 2023). It also

supports NFT-based property representation, fractional ownership, tokenized investment, security-token participation, and marketplace coordination (Joshi and Choudhury 2022; Abualhamayl et al. 2025; Desai et al. 2024; Kranz et al. 2026).

Tokenization nevertheless augments rather than replaces institutions. Because real-estate rights are legally constructed and institutionally recognized, a token's practical value depends on the enforceability and registry recognition of its underlying right, regulatory compliance, and user trust. Highly regulated markets additionally connect property transactions to official registration, documentation, financial oversight, identity verification, valuation, and public governance.

Despite growing research on blockchain in real estate, an integration gap remains. Studies and reviews predominantly examine components such as land-registry integrity, property representation through NFTs, fractional investment, smart-contract automation, institutional adoption, or broader real-world-asset tokenization (Saari et al. 2022a; Ansah et al.

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2023; Jaouhari et al. 2025; Luo et al. 2026). Their largely separate accounts of architecture, property rights, certification, compliance, stakeholders, and acceptance do not provide an integrated evidence structure showing how token representation, legally recognized rights, official certification, identity and compliance controls, marketplace processes, and user perceptions align in regulated markets. This limits system comparison and translation into implementable governance and design requirements.

Accordingly, this study treats tokenization as an end-to-end regulated marketplace problem. It connects a multi-dimensional classification with legal and institutional constraints, exploratory regulated-market perceptions, and a jurisdiction-gated governance framework. Its novelty lies in an evidence-traceable, design-oriented synthesis; token-standard design and deployed-market validation represent distinct research scopes.

The mixed-method design combines a PRISMA-based systematic review of 63 primary studies (2020–May 2026) with an exploratory respondent-perception survey of 153 participants in a highly regulated market. These sources support gap identification and framework development, while a separate institutional-evidence comparison assesses applicability against documented initiatives.

Accordingly, the study has four objectives:

1. to systematically identify and classify blockchain-based real-estate-tokenization research across architectural, ownership, transactional, certification, stakeholder, technological, and adoption dimensions;
2. to synthesize the technical, legal, institutional, governance, identity, privacy, valuation, interoperability, scalability, pre-deployment assurance, dynamic security, and long-term cryptographic requirements shaping regulated deployment;
3. to examine respondents' perceptions of selected blockchain-enabled real-estate services in a highly regulated market through exploratory, sample-specific interpretation; and
4. to translate the combined evidence into a socio-technical governance framework, jurisdiction-gated adoption pathway, and evidence-linked design requirements for regulated tokenization marketplaces, and to assess their applicability against documented institutional initiatives.

The four contributions correspond to these objectives. First, a multidimensional taxonomy compares architectures, represented rights, transactions, certification, stakeholders, technologies, and adoption barriers. Second, a cross-dimensional synthesis shows that regulated deployment requires alignment among technical mechanisms, recognized property rights, institutional authority, identity and compliance, valuation, privacy, interoperability, pre-

deployment verification, continuous monitoring, and cryptographic resilience. Third, exploratory regulated-market evidence identifies a coherent general orientation and distinguishes recognition of particular capabilities from endorsement of their wider adoption consequences. It provides contextual adoption conditions and hypotheses for confirmatory research. Fourth, the combined evidence yields an evidence-linked socio-technical framework comprising a jurisdiction-gated pathway and twelve design requirements, with institutional applicability assessed against documented initiatives.

The remainder of the paper presents related work (Section 2); the research framework, taxonomy, and questions (Section 3); the review, survey, and institutional-evidence methodology (Section 4); review and respondent-perception results (Sections 5–6); cross-dimensional synthesis and gaps (Sections 7–8); the governance framework, design requirements, and institutional alignment (Section 9); and validity considerations and conclusions (Sections 10–11).

2 Motivation and related work

The motivation for this study arises from the fragmented state of existing review and survey-based research on blockchain in real estate. Prior reviews have established that blockchain can support land administration, property transactions, leasing, investment, identity, and institutional trust. However, these studies usually examine separate parts of the ecosystem rather than real-estate tokenization as an integrated regulated marketplace from an information systems perspective.

Broad reviews of blockchain in real estate provide the foundation for this work. Saari et al. (2022a) reviewed blockchain's promise for the real-estate sector and classified prior studies across land administration, property transactions, investment, leasing, and administration. In a related study, Saari et al. (2022b) examined recent developments and empirical applications of blockchain in real estate. These reviews are valuable because they show the breadth of blockchain-real-estate research and the uneven maturity of the field. However, their scope is broader than tokenization and does not provide a dedicated taxonomy of real-estate tokenization systems in terms of ownership models, transaction types, tokenization mechanisms, certification approaches, stakeholder structures, technology models, and adoption barriers.

A second group of reviews focuses on land administration and registry modernization. Mezquita et al. (2022) surveyed blockchain-based land registry platforms and highlighted transparency, cost reduction, execution time, and e-government transformation. Ansah et al. (2023) systematically reviewed institutional success factors for blockchain-based land administration and showed that successful adop-

tion depends on regulatory, organizational, and cultural-cognitive factors. Riaz and Ikram (2023) reviewed blockchain as a response to land registry problems such as forgery, document loss, and slow ownership-transfer procedures. These studies are directly relevant to the registry and institutional dimensions of real-estate digitalization. Nevertheless, they remain centered on land administration and do not fully address tokenized ownership, NFTs, fractional investment, security-token models, marketplace workflows, or respondent perceptions of real-estate tokenization.

A third group examines specific functions or adoption dimensions. Jain et al. (2024) surveyed blockchain for rental lease management, focusing on lease transparency, payment handling, identity verification, and dispute reduction. Shuaib et al. (2022) compared identity models for blockchain-based land registry systems and emphasized the need to connect blockchain wallets with legally recognized identities. More recent empirical adoption studies move beyond technical feasibility toward user acceptance and psychological adoption factors. Yeoh et al. (2024) examined the acceptance of blockchain by real-estate buyers and sellers, designing a research model combining UTAUT and the technology readiness index, collecting data from 301 real-estate stakeholders, and finding that psychological rather than technological factors drive adoption. Yang et al. (2025) extended this line of inquiry through an integrated TAM and task-technology-fit perspective on real-estate blockchain adoption. These studies are important contributions because they establish that adoption in real-estate blockchain contexts is governed by trust, psychological readiness, and facilitating conditions rather than by technical capability alone (Davis 1989; Venkatesh et al. 2003). However, they do not focus specifically on tokenized real-estate marketplaces, fractional ownership, token-based investment, or the socio-technical governance conditions required for regulated-market adoption.

More recently, Jaouhari et al. (2025) provided the closest review to the present work by examining real-estate tokenization through a Technology–Organization–Environment perspective. Their review identifies applications, drivers, barriers, and future directions for real-estate tokenization. However, it mainly frames tokenization as a property-investment and adoption phenomenon. The present paper differs by providing a taxonomy-based systematic review focused on blockchain-based real-estate tokenization as a regulated digital marketplace. It classifies the literature across architectural families, ownership models, transaction types, tokenization models, certification mechanisms, stakeholder structures, technology models, and adoption barriers. It also connects the literature synthesis with survey-based respondent-perception evidence from highly regulated real-estate markets and derives a socio-technical governance framework and design requirements for regulated real-estate tokenization marketplaces.

At the wider real-world-asset level, Luo et al. (2026) systematized real-world asset tokenization across concepts, architectures, lifecycle structures, and legal interoperability. Chemla and Tinn (2026) further argue that tokenization does not create value merely by making assets transferable, fractional, or digitally scarce, but only when digital ownership records support economic mechanisms such as provenance, retained ownership value, separated rights, membership benefits, or post-sale incentives. These works are useful for positioning real-estate tokenization within the broader RWA-tokenization landscape. However, they are not domain-specific reviews of real-estate tokenization and do not examine property-specific registry integration, ownership-transfer logic, certification, valuation, stakeholder governance, or the socio-technical conditions required for regulated marketplace adoption.

Table 1 summarizes the positioning of the present study against existing review and survey-based work.

The comparison establishes four interdependent gaps. The first is a *classification gap*: existing reviews do not provide a common structure for comparing system architecture, represented rights, transaction types, tokenization and certification mechanisms, stakeholder responsibilities, implementation technologies, and adoption barriers. The second is a *socio-technical integration gap*: registry, investment, adoption, and RWA studies do not sufficiently explain how these dimensions must operate together in an end-to-end regulated marketplace. The third is an *adoption-context gap*: tokenization-specific perceptions in a highly regulated, institutionally anchored real-estate setting remain under-examined. The fourth is an *evidence-to-design translation gap*: the literature does not provide a traceable route from observed system patterns and deployment barriers to staged governance and design requirements. These gaps define the study's contribution: a taxonomy for structured comparison, a cross-dimensional synthesis, exploratory regulated-market perception evidence, and an evidence-linked socio-technical governance framework. This positioning moves beyond summarizing blockchain applications and clarifies how the reviewed evidence can inform the design and governance of regulated real-estate-tokenization marketplaces.

3 Research framework and study objectives

The present study therefore develops a taxonomy-based research framework for understanding blockchain-based real-estate tokenization as a regulated marketplace ecosystem. Real-estate tokenization is treated not merely as a method for recording property data on-chain, but as a socio-technical, commercial, and regulatory model connecting property rights, digital assets, transaction coordination, certification mechanisms, stakeholder responsibilities, compli-

Table 1 Positioning of the present review against existing review and survey-based studies

Study stream	Main focus	Remaining gap addressed by this paper
Broad blockchain-real-estate reviews (Saari et al. 2022a, b)	Applications across land administration, transactions, investment, leasing, and administration.	This paper adds a tokenization-specific taxonomy covering token models, certification, stakeholders, and regulated-marketplace requirements.
Land administration and registry reviews (Mezquita et al. 2022; Ansah et al. 2023; Riaz and Ikram 2023)	Land-registry platforms, institutional adoption factors, and land-record modernization.	This paper extends registry-focused evidence to NFTs, fractional ownership, tokenized investment, marketplace workflows, and respondent perceptions.
Adoption studies (Yeoh et al. 2024; Yang et al. 2025)	Acceptance among real-estate buyers, sellers, and users through TAM, UTAUT, and task-technology-fit models.	This paper connects adoption factors with an integrated tokenization ecosystem, socio-technical governance, and jurisdiction-gated design.
Function-specific studies (Jain et al. 2024; Shuaib et al. 2022)	Rental management and identity models for blockchain land registries.	This paper integrates function-level evidence with marketplace governance and regulated-adoption requirements.
Real-estate tokenization review (Jaouhari et al. 2025)	Applications, drivers, barriers, and future directions through a TOE perspective.	This paper extends the closest review with a system taxonomy, governance architecture, and design requirements.
RWA tokenization systematization and theory (Luo et al. 2026; Chemla and Tinn 2026)	Broader RWA concepts, architectures, legal interoperability, and economic-value mechanisms.	This paper adds property-specific ownership, registry integration, certification, valuation, inspection, and regulated-market governance.

ance requirements, and user trust. The taxonomy constitutes a review-derived, design-oriented classification artifact that supports structured comparison and provides the analytical foundation for the subsequent gap synthesis. Its dimensions were derived from related reviews before corpus formation and refined iteratively as primary studies were coded, following established SLR-based design-science practice (Kitchenham et al. 2009).

The taxonomy was constructed from the structure of the reviewed domain rather than imposed as an abstract classification. Three evidence layers informed its dimensions. First, broad real-estate blockchain reviews show the field is distributed across land administration, property transactions, investment, leasing, and administration (Saari et al. 2022a, b). Land-administration and registry reviews further show that blockchain adoption depends on institutional authority, legal recognition, registry integration, and public-sector governance (Ansah et al. 2023; Mezquita et al. 2022; Riaz and Ikram 2023). More recent tokenization and RWA systematization studies show that tokenization must be examined through ownership representation, asset-right linkage, token standards, legal interoperability, and market adoption (Jaouhari et al. 2025; Luo et al. 2026; Chemla and Tinn 2026). These sources establish that blockchain-based real-estate systems must be examined not only as technical systems but also as institutional, transactional, and marketplace systems.

Second, the primary studies included in the review revealed recurring system patterns that a single category could not capture. Registry-centric studies focus on tamper-

evident records, deed management, and registry updates (Banerjee et al. 2022; Yadav and Kushwaha 2021; Mallya et al. 2025). NFT-based studies examine unique property representation, digital twins, and token-enabled transfer (Sharma et al. 2024; Vivekrabinson et al. 2023; Jayakumar et al. 2025; Mitrea et al. 2025; Hassan et al. 2024). Fractional ownership and tokenized investment studies focus on shared participation, liquidity, investor access, security-token offerings, and secondary-market dynamics (Abualhamayl et al. 2025; Desai et al. 2024; Avci and Erzurumlu 2023; Joshi and Choudhury 2022; Kreppmeier et al. 2023; Swinkels 2023; Serogina and Prokopiev 2025). Security-token and DeFi-oriented studies extend this stream by examining SPV structures, token standards, KYC/AML, whitelisting, batch transfers, and DeFi integration (Kranz et al. 2026; Henker et al. 2023; Zhang et al. 2024b). Rental, leasing, and mortgage studies extend blockchain use toward usage rights and financial workflows (Jain et al. 2024; Wang et al. 2022; Junaib et al. 2024). Hybrid studies combine certification, escrow, verification, regulatory approval, identity assurance, and marketplace coordination (Exter and Radosavljevic 2023; Jamshed et al. 2025; Jeong and Ahn 2021; Henriques and Correia 2025; Sabah 2022; Le et al. 2025; Preethi et al. 2026).

Third, deployment concerns from identity, adoption, privacy, and long-term security literature informed the adoption-barrier and technology-readiness dimensions. Blockchain wallets cannot by themselves establish legal identity, ownership authority, or transaction eligibility (Shuaib et al. 2022). Empirical real-estate blockchain studies identify perceived

usefulness, ease of use, trust, and facilitating conditions as relevant acceptance factors (Yeoh et al. 2024; Yang et al. 2025). Post-quantum blockchain security research further highlights the need for long-term cryptographic resilience in systems whose records must remain valid for extended periods (Fernández-Caramés and Fraga-Lamas 2020).

Table 2 summarizes the construction process from source evidence to taxonomy dimensions.

The final taxonomy organizes reviewed studies across eight interdependent dimensions: architectural family, ownership model, transaction type, tokenization model, certification mechanism, stakeholder structure, technology model, and adoption barrier. These dimensions are interdependent because, for example, a fractional ownership model implies investment-related transactions, stronger compliance requirements, valuation mechanisms, and investor-governance rules. An NFT-based model may be technically suitable for representing a unique property, but its value depends on whether the NFT represents legal title, a digital certificate, a fractional claim, or only a pointer to an off-chain record. Security-token and DeFi-oriented models further introduce concerns related to SPV structures, investor onboarding, whitelisting, token standards, cross-platform interoperability, and regulatory supervision.

Figure 1 illustrates the taxonomy.

The taxonomy provides the analytical basis for the following research questions:

- RQ1.** What architectural families characterize blockchain-based real-estate tokenization systems?
- RQ2.** Which ownership, tokenization, certification, and stakeholder models dominate existing real-estate tokenization research?
- RQ3.** What transaction types are supported by current systems, including selling, buying, renting, leasing, financing, investment, resale, and registry update?
- RQ4.** What technologies are used to implement real-estate tokenization, including NFTs, fungible tokens, security-token standards, smart contracts, off-chain storage, oracles, privacy mechanisms, DeFi integration, and long-term security mechanisms?
- RQ5.** What technical, legal, governance, privacy, identity, valuation, scalability, verification, interoperability, and regulatory gaps limit adoption?
- RQ6.** How do surveyed respondents in a highly regulated real-estate market perceive blockchain-based real-estate tokenization in terms of trust, usability, investment intention, risk perception, fractional ownership, token-based transactions, and regulatory confidence?

Table 2 Construction process of the taxonomy

Evidence source	Observed pattern	Taxonomy dimensions informed
Broad real-estate blockchain reviews	Research spans land administration, property transactions, investment, leasing, and administration.	Architectural family; transaction type; adoption barrier.
Land-administration and registry reviews	Adoption links legal recognition, official registries, institutional authority, and public-sector governance.	Certification mechanism; stakeholder structure; regulatory and governance barriers.
Tokenization and RWA systematization studies	Tokenization links asset rights, legal interoperability, ownership representation, liquidity, and market adoption.	Ownership model; tokenization model; certification mechanism; adoption barrier.
Registry-centric primary studies	Blockchain provides a tamper-evident layer for deeds, ownership records, document hashes, and registry updates.	Architectural family; certification mechanism; transaction type.
NFT and property-representation studies	NFTs or related mechanisms represent properties, certificates, and digital twins.	Ownership model; tokenization model; certification mechanism; technology model.
Fractional investment and security-token studies	Tokenization supports shared ownership, investment, liquidity, secondary trading, and investor access.	Ownership model; transaction type; tokenization model; stakeholder structure; adoption barrier.
DeFi-oriented and regulated marketplace studies	SPVs, token standards, KYC/AML, whitelisting, onboarding, batch transfers, and DeFi integration.	Architectural family; technology model; stakeholder structure; certification mechanism; adoption barrier.
Rental, leasing, mortgage, and workflow studies	Usage rights, payments, lease records, mortgage workflows, escrow, and conditional execution.	Transaction type; stakeholder structure; technology model.
Identity, adoption, privacy, and long-term security	Deployment links identity assurance, KYC/AML, privacy, usability, trust, and long-term cryptographic resilience.	Technology model; stakeholder structure; adoption barrier.

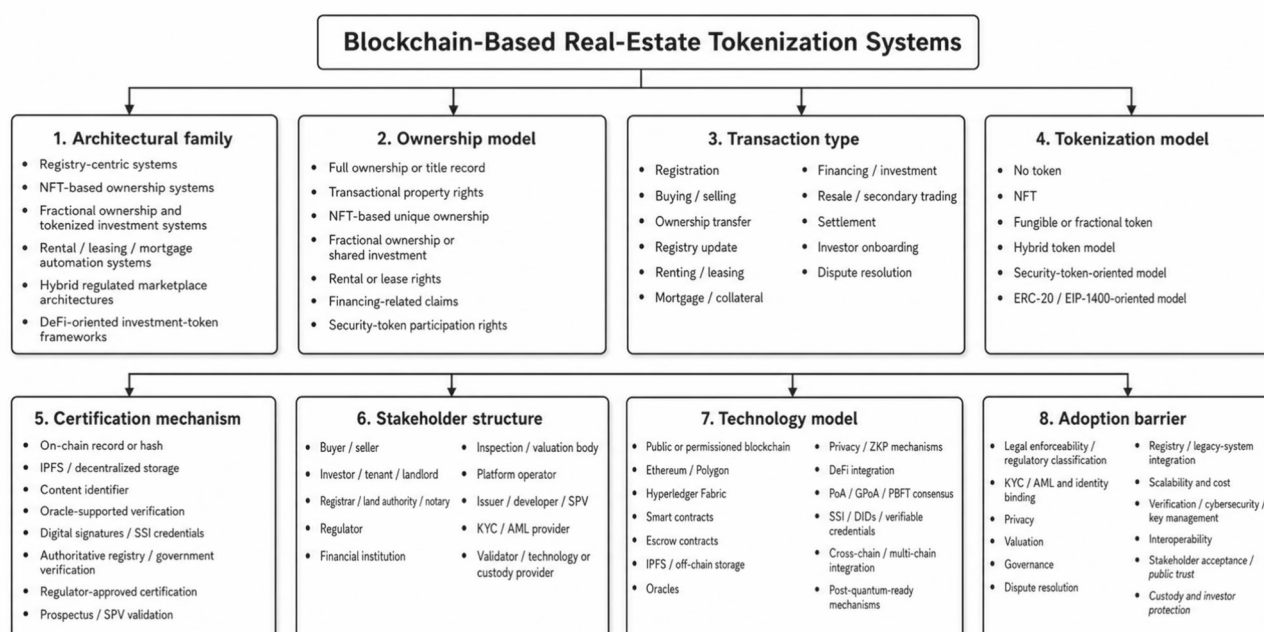


Fig. 1 Taxonomy of blockchain-based real-estate tokenization systems used in the systematic review

RQ7. What socio-technical governance framework and design requirements can support blockchain-based real-estate tokenization in regulated markets?

Together, the research questions establish a traceable sequence from evidence classification to gap synthesis, contextual perception analysis, and framework development. RQ1–RQ5 address the classification and integration gaps through the systematic review; RQ6 addresses the adoption-context gap through the exploratory perception survey; and RQ7 addresses the evidence-to-design translation gap through the cross-dimensional synthesis and socio-technical governance framework. The resulting contribution is therefore the connected evidence chain across these components, not any component considered in isolation.

4 Research methodology

The methodology provides an evidence-traceable sequence from research design and source identification through study selection, extraction, appraisal, survey analysis, synthesis, and institutional applicability assessment for RQ1–RQ7.

4.1 Research design

This study adopted a multi-stage mixed-method research design combining a systematic literature review, an exploratory perception survey, cross-dimensional synthesis, and an institutional applicability assessment. The review supplied

the principal evidence base for classifying and analysing blockchain-based real-estate tokenization research. The survey supplied descriptive evidence on blockchain documentation, integrated services, investment, and tokenized transactions, together with respondents' stated advantages and concerns. The final synthesis integrated both evidence streams to derive the research gaps and the governance-aware framework. The institutional assessment then examined how documented initiatives map to the framework stages and design requirements. RQ1–RQ5 were addressed through the systematic review, RQ6 through the exploratory perception survey, and RQ7 through the cross-dimensional synthesis and institutional applicability assessment.

The review followed established systematic-review and mapping-study guidance for planning, retrieval, study selection, data extraction, quality appraisal, coding, and synthesis (Kitchenham et al. 2009; Petersen et al. 2015). PRISMA 2020 reporting principles were used to document the identification, screening, retrieval, and inclusion flow (Page et al. 2021).

4.2 Data sources and search strategy

The search strategy was designed to capture blockchain-based real-estate tokenization as a multidisciplinary and rapidly evolving research area. The literature uses varied terminology, with relevant studies indexed under blockchain real estate, land registry, smart contracts, property tokenization, NFT-based ownership, fractional ownership, real-world asset tokenization, security tokens, DeFi integration, and digital asset marketplaces. The objective was to retrieve studies

contributing to at least one taxonomy dimension defined in Section 3.

The search covered studies published between 2020 and May 2026, a period selected because real-estate tokenization research expanded substantially with the emergence of NFT-based property representations, fractional investment systems, security-token models, DeFi-oriented frameworks, and broader real-world-asset tokenization. Four complementary routes were used to reduce database bias, capture recent work, and identify studies missed by keyword-based searches. All candidate records were screened using the same inclusion and exclusion criteria regardless of route.

The inclusion criteria were:

- IC1.** Published between 2020 and May 2026.
- IC2.** Focuses on blockchain, DLT, or smart contracts in real estate, land registry, property transfer, leasing, financing, tokenization, or digital asset marketplaces.
- IC3.** Provides a technical, architectural, methodological, marketplace, or regulatory contribution.
- IC4.** Contains sufficient detail to support coding of at least one taxonomy dimension.
- IC5.** Written in English and available as a full-text peer-reviewed journal article, conference paper, or scholarly book chapter.

The exclusion criteria were:

- EC1.** Blockchain or cryptocurrency papers without substantive connection to real estate, land administration, or tokenized asset markets.
- EC2.** Purely legal, economic, or opinion-based papers without transferable technical, marketplace, or regulatory content.
- EC3.** Duplicate or substantially overlapping versions of the same study.
- EC4.** Abstracts, posters, editorials, theses, dissertations, and non-peer-reviewed materials.
- EC5.** Studies bound to a single local legal arrangement without transferable technical, architectural, marketplace, governance, or regulatory evidence.
- EC6.** Studies mentioning real estate only as an illustrative example.
- EC7.** Secondary reviews and systematization papers (usable in related work but excluded from the primary corpus).

Full-text availability and verifiable peer-reviewed publication status were required for inclusion. A DOI was not required when the full text and publication status could be verified. Country-specific studies were retained when they provided transferable technical, architectural, marketplace, governance, or regulatory evidence. Secondary reviews and systematization papers informed the related-work analysis

and citation-chaining route; the primary-corpus count comprises primary peer-reviewed studies.

The review covered studies published from 2020 to May 2026. The same core search expression was used in both database-search phases and entered in Elicit; the same search concepts and study-selection criteria guided citation chaining and the recent-literature scan.

The first database-search phase was conducted from September to December 2025 across Scopus, Web of Science, IEEE Xplore, ScienceDirect, ACM Digital Library, SpringerLink, and relevant publisher databases. It returned 1,840 records. After identifying 205 duplicate records, 1,635 unique records were screened. Title and abstract screening excluded 1,591 records: 1,395 outside the real-estate scope, 123 survey or review papers, 28 country-specific records without transferable evidence, and 45 other out-of-scope records. Forty-four reports advanced to retrieval; 2 reports were unavailable, leaving 42 full texts for eligibility assessment. One report was excluded at the framework criterion, and 41 primary studies were retained from Phase 1.

The second database-search phase was conducted from March to May 2026 using the same databases and search expression. It returned 1,211 records. Within-phase deduplication removed 352 records, leaving 859 unique records for title and abstract screening. Screening retained 47 reports for retrieval, of which 44 were available for full-text eligibility assessment. One report fell outside the framework criterion, and one substantially overlapping version was removed under EC3 during study-level matching, yielding 42 distinct primary studies. Cross-phase matching showed that 41 of these studies were already represented in Phase 1 and one study was newly added, producing a 42-study database-search set.

Duplicate removal was performed independently within each database phase. The duplicate proportions were 11.1% in Phase 1 (205 of 1,840 records) and 29.1% in Phase 2 (352 of 1,211 records). Although both phases used the same search expression and sources, duplicate prevalence depends on the overlap within the combined result set returned during each retrieval; the second phase contained a greater proportion of repeated records. Cross-phase study matching was conducted separately after eligibility assessment and was therefore distinct from these within-phase duplicate counts.

The core search expression was:

("blockchain" OR "distributed ledger" OR "DLT") AND ("real estate" OR "property" OR "land registry" OR "land title" OR "housing") AND ("tokenization" OR "tokenisation" OR "NFT" OR "fractional ownership" OR "smart contract" OR "digital asset marketplace")

A supplementary expression captured token standards, off-chain storage, and long-term security concerns:

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("real estate tokenization" OR "property tokenization" OR "tokenized real estate")AND("NFT" OR "ERC-721" OR "ERC-20" OR "ERC-1155" OR "EIP-1400" OR "security token" OR "smart contract" OR "IPFS" OR "post-quantum")
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The second route was backward and forward citation chaining applied to retrieved studies, recent systematic reviews, and systematization papers, to capture studies missed due to terminology variation across blockchain real estate, property tokenization, land registry, NFT property, security-token offering, and DeFi integration. The third route was an Elicit-assisted supplementary search used as a completeness check; all candidate records were manually verified using the same study-selection criteria. The fourth route was an AI-assisted exploratory scan of recent preprints, early-access publications, and institutional literature, including SSRN, arXiv, policy reports, and publisher pages, to identify very recent developments. Peer-reviewed studies satisfying the selection criteria entered the primary corpus, while the remaining sources informed contextual discussion.

Citation chaining identified 22 candidate records. De-duplication and screening identified 4 duplicate or overlapping records, 12 primary studies for inclusion, 4 contextual or related-work records, and 2 records outside the eligibility criteria. The Elicit query returned 500 records, of which 17 remained after topic and scope filtering. Manual verification identified 8 primary studies for inclusion, 2 duplicate or overlapping records, 3 contextual or related-work records, 2 records outside the eligibility criteria, and 2 records without verifiable metadata or accessible full text. The recent-source

scan contributed 1 peer-reviewed real-estate-specific study; its other outputs supported the contextual analysis. Together, the three supplementary routes contributed 21 primary studies.

A separate purposive institutional-evidence route supported the contextual comparison and framework-applicability assessment. Official government, land-administration, regulatory, and intergovernmental publications were retrieved using terms covering blockchain land registration, real-estate tokenization, fractional investment, and secondary-market operation. An initiative entered the six-case stage mapping when authoritative material identified the responsible institution and disclosed sufficient functional or architectural detail to assess multiple framework stages or design requirements, while narrower disclosures informed the contextual discussion. This institutional evidence formed a distinct contextual set, while the 63-study PRISMA corpus, extraction matrix, quality appraisal, and evidence-maturity analysis remained confined to eligible peer-reviewed primary studies.

4.3 Study selection and corpus formation

The two database phases and three supplementary routes were consolidated through record-level de-duplication, full-text screening, study-level matching, and application of the IC/EC criteria. Table 3 reports the dates, data sources, and study contribution of each route.

Figure 2 presents study identification, screening, and inclusion across the two database phases and three supplementary routes. Cross-phase matching produced 42 database studies, while citation chaining, the Elicit-assisted search, and the recent-source scan contributed 12, 8, and 1 studies, respectively. These routes yielded the 63-study corpus summarized in Table 3 and shown in the figure.

The corpus spanned registry-centric systems, NFT-based property representations, fractional ownership and tokenized

Table 3 Search routes and study contributions. Bold indicates final/aggregate total values

Search route	Date	Databases or sources	Final output
Database search — Phase 1	Sep–Dec 2025	Scopus; Web of Science; IEEE Xplore; ScienceDirect; ACM Digital Library; SpringerLink; relevant publisher databases	41 studies retained
Database search — Phase 2	Mar–May 2026	Same databases and search expression as Phase 1	1 new study; database-search set after cross-phase matching = 42
Backward and forward citation chaining	May 2026	Reference lists and citing records of retrieved studies, recent systematic reviews, and systematization papers	12 studies added
Elicit-assisted supplementary search	May 2026	Elicit default scholarly sources using the same query	8 studies added
AI-assisted recent/preprint scan	May 2026	SSRN; arXiv; early-access publisher pages; institutional literature	1 peer-reviewed study added
Final primary-study corpus	May 2026	Two database phases and three supplementary routes	63 studies: database searches 42; citation chaining 12; Elicit 8; recent scan 1

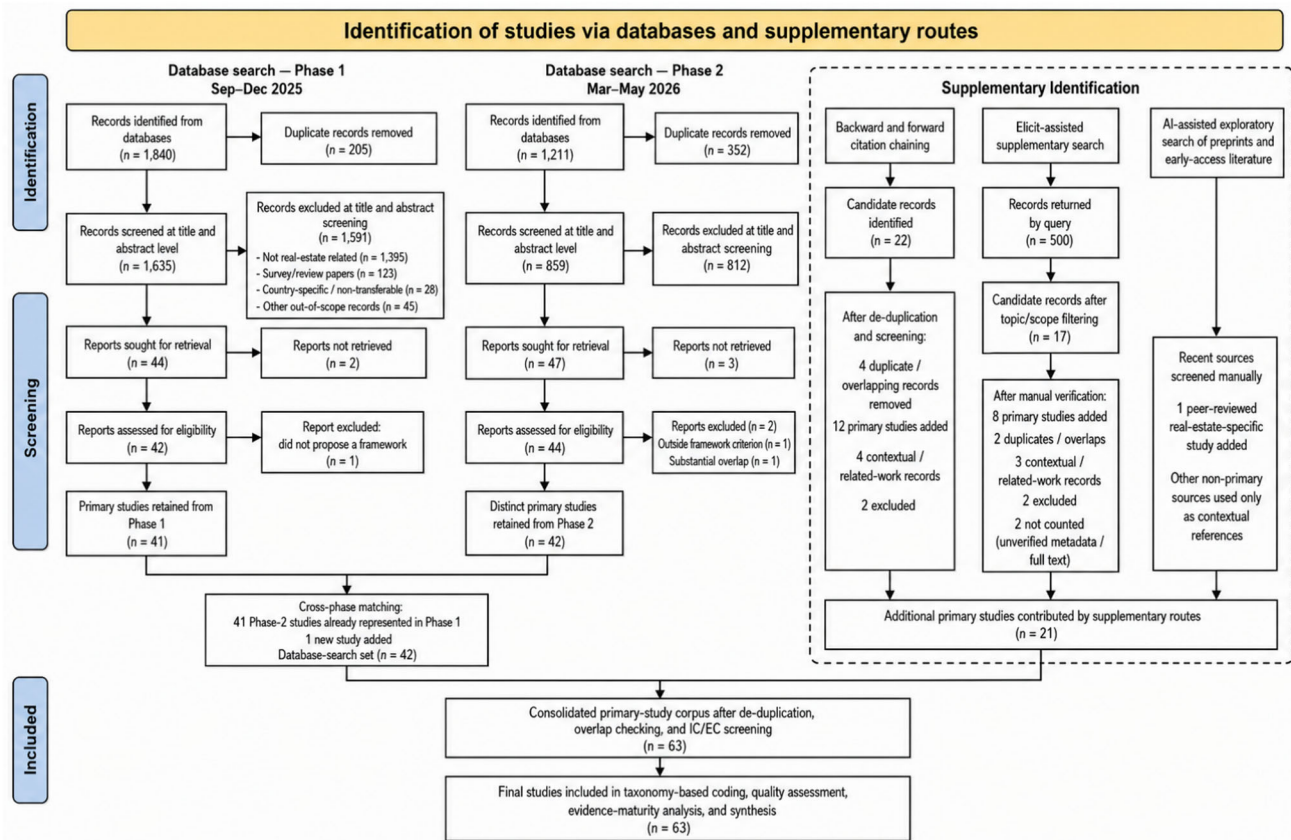


Fig. 2 PRISMA flow for study identification, screening, and corpus formation

investment models, rental and mortgage workflow systems, security-token and DeFi-oriented investment frameworks, and hybrid regulated marketplace architectures — providing sufficient breadth to support taxonomy-based synthesis across all eight dimensions.

4.4 Data extraction, quality assessment, and evidence maturity

The 63 studies entered a 20-field extraction matrix grouped into descriptive, taxonomy-based, and interpretive categories. Descriptive fields recorded publication year, venue, method, application focus, evaluation, and limitations. Taxonomy fields covered architecture, ownership, transactions, tokenization, certification, stakeholders, technology, and adoption barriers; interpretive fields covered maturity, regulation, identity/KYC, privacy, security assumptions, and long-term security readiness. The matrix supported up to $63 \times 20 = 1,260$ entries, with unreported fields coded *not reported*. Each study received one dominant architectural family and primary maturity level, while secondary characteristics remained in the extraction notes.

Dominant architectural classifications were: 21 registry-centric and deed-management systems, 14 fractional owner-

ship and tokenized investment systems, 13 hybrid regulated marketplace architectures, 8 NFT-based ownership systems, 4 rental/leasing/mortgage automation systems, and 3 DeFi-oriented investment-token frameworks. Secondary architectural features remained in the notes and were excluded from dominant-category counts.

Methodological quality was appraised using five criteria adapted from systematic-review and mapping-study guidance (Kitchenham et al. 2009; Petersen et al. 2015): clarity of the research objective; relevance to blockchain-based real-estate tokenization; clarity of the proposed architecture, framework, model, or method; adequacy of evaluation or supporting evidence; and explicit discussion of limitations, risks, or assumptions. Scores of 0, 0.5, or 1 yielded a maximum of 5 per study and 315 criterion-level judgments. Both authors applied the rubric jointly and resolved differences by consensus. The complete scoring matrix is provided in the supplementary files. The appraisal classified 32 studies as higher quality (4.0–5.0), 31 as moderate (2.5–3.5), and none as lower quality (0–2.0), with a mean of 3.99/5.

Evidence maturity was assigned as one primary level per study: 27 prototype or proof-of-concept, 22 simulation or experimental evaluation, 5 conceptual proposal, 4 architecture or framework model, 3 empirical or expert-based

evaluation, 2 case or scenario evaluation, and no deployed or operational system.

Table 4 summarizes aggregate coding outcomes.

The 48 studies concentrated in registry-centric, fractional or tokenized-investment, and hybrid-marketplace families show movement beyond land-record digitization, although prototype and simulation/experimental evidence remains dominant and the corpus contains no operational deployment. Appendix A details the fields, categories, quality rubric, and maturity rules.

4.5 Analytical approach

The analysis linked descriptive mapping, taxonomy coding, quality and maturity appraisal, survey analysis, and cross-dimensional synthesis. Likert responses were summarized as agreement (4–5), neutral (3), and disagreement (1–2). Aggregate response consistency was assessed using Cronbach's alpha with a 2,000-resample bootstrap confidence interval, standardized alpha, corrected item–total correlations, and alpha-if-item-deleted. An unrotated single-factor principal-component assessment provided a basic common-method check of the general response disposition and shared questionnaire method.

Spearman correlations used all 153 complete response rows, with Holm adjustment across 28 pairwise tests. A sensitivity assessment repeated the alpha, single-factor, and correlation analyses for differentiated response profiles. Open answers were coded literally, with multiple explicitly stated ideas retained. Respondent-level correspondence then compared each explicit open-response component with the

complete closed item. The survey analyses were interpreted exploratorily, and cross-dimensional comparison derived the research gaps and governance requirements.

The manuscript and appendix document the questions, search expressions, sources, dates, criteria, screening flow, extraction fields, quality rubric, maturity rules, and synthesis procedure.

4.6 Survey-based respondent perceptions

The survey extends the review's adoption perspective through exploratory evidence on blockchain documentation, integrated services, investment, and tokenized transactions among respondents in Saudi Arabia. Interpretation remains sample-specific and informs design requirements and future confirmatory research.

The two-part questionnaire comprised eight required five-point Likert items, reproduced in full English translation in Table 10. Two open prompts then requested the most important additional advantage and concern, explicitly directing respondents toward considerations not mentioned in the closed items. Mandatory closed items yielded complete submissions. Two confidential domain experts affiliated with a regulatory authority reviewed the instrument for clarity and contextual appropriateness.

Distribution in Saudi Arabia used social-media channels and targeted contact lists spanning university faculty, real-estate professionals, state-agency contacts, and adults (18+) from varied backgrounds. This highly regulated, registry-connected setting provides broad descriptive perceptions rather than a probability-based national sample.

Table 4 Aggregate coding and quality-appraisal outcomes

Coding dimension	Dominant category	Count	Share
Architectural family	Registry-centric and deed-management systems	21	33.3%
	NFT-based ownership systems	8	12.7%
	Fractional ownership and tokenized investment systems	14	22.2%
	Rental/leasing/mortgage automation systems	4	6.3%
	Hybrid regulated marketplace architectures	13	20.6%
	DeFi-oriented investment-token frameworks	3	4.8%
Evidence maturity	Conceptual proposal	5	7.9%
	Architecture or framework model	4	6.3%
	Prototype or proof of concept	27	42.9%
	Simulation or experimental evaluation	22	34.9%
	Case study or scenario evaluation	2	3.2%
	Empirical or expert-based evaluation	3	4.8%
	Deployed or operational system	0	0.0%
Quality-score band (63-study appraisal)	Higher-quality evidence, 4.0–5.0	32	50.8%
	Moderate-quality evidence, 2.5–3.5	31	49.2%
	Lower-quality evidence, below 2.5	0	0.0%

The dataset comprised 153 valid responses; scores 4–5 denoted agreement, 3 neutrality, and 1–2 disagreement. Recruitment and the focus on perception variables define aggregate, sample-specific findings subject to self-selection. Future replication can add age, education, experience, and role comparisons (Section 10, T4).

4.7 Cross-dimensional synthesis

The final methodological step integrated the review and survey evidence in three stages. Review results were organized by taxonomy dimensions and research questions across the six architectural families. Survey results identified the general response orientation and the assurance conditions accompanying adoption judgments. Comparison of the two streams then derived evidence-linked gaps, future directions, governance stages, and design requirements. A subsequent framework-applicability assessment mapped the institutional evidence set to the five stages and twelve design requirements using explicitly documented functions. Table 5 summarizes this analytical sequence.

5 Results of the systematic review

The 63-study corpus spans blockchain-based land registries, property-transfer systems, NFT-based ownership, digital twins, fractional ownership models, security-token and DeFi-oriented investment frameworks, rental and leasing workflows, mortgage-related processes, and hybrid marketplace infrastructures, providing the evidence for RQ1–RQ5 across the eight taxonomy dimensions defined in Section 3. The publication distribution comprised 4 studies in 2020, 5 in 2021, 10 in 2022, a peak of 18 in 2023, 14 in 2024, 10 in 2025,

and 2 in 2026. The 2024–2026 studies continued the shift toward tokenized investment, security-token standards, and regulated marketplace design, indicating a gradual evolution from land-record digitization toward marketplace-oriented tokenization models.

5.1 RQ1: Architectural families

The review identifies six architectural families differing in purpose, legal assumptions, transaction scope, and maturity.

Registry-centric systems use blockchain as a tamper-evident infrastructure for land records, deeds, ownership history, and document verification (Humdullah et al. 2021; Yadav and Kushwaha 2021; Ncube et al. 2022; Banerjee et al. 2022; Borse et al. 2022; Yadav et al. 2023; Mehta et al. 2023; Sharma et al. 2023; Prabu et al. 2023; Singh et al. 2023; Ozcelik 2024; Zeid et al. 2024; Gadagkar and Begum 2024; Poliseti et al. 2025; Merchant et al. 2022; Stefanović et al. 2022; Henriques and Correia 2025; Soner et al. 2021; Thakur et al. 2020; Zhang et al. 2024a; Mallya et al. 2025). They provide the foundational layer without which tokenized property markets cannot function reliably.

NFT-based ownership systems represent a property, certificate, or digital twin as a unique digital asset using ERC-721-like logic (Rana et al. 2022; Aliti et al. 2023; Vivekrabinson et al. 2023; Sharma et al. 2024; Hassan et al. 2024; Jayakumar et al. 2025; Mitrea et al. 2025; Arcenegui et al. 2023). The legal meaning of the NFT remains underdeveloped in many studies — it may represent legal ownership, a digital certificate, a proof of provenance, or merely an access mechanism linked to off-chain documentation.

Fractional ownership and tokenized investment systems divide property value or participation rights into tokenized shares, addressing liquidity, accessibility, and diversification

Table 5 Cross-dimensional synthesis logic

Evidence source	Analytical role	Output in the paper
Systematic review corpus	Classifies 63 primary studies by taxonomy dimensions; answers RQ1–RQ5.	Architectural families, ownership models, transaction types, tokenization models, certification mechanisms, stakeholder structures, technology models, and adoption barriers.
Quality and evidence-maturity assessment	Supports interpretation according to evidence strength and maturity.	Distinction between conceptual, architectural, prototype-based, simulation-based, empirical, expert-based, and deployed evidence.
Survey-based respondent-perception component	Examines closed responses, response consistency, open themes, and respondent-level correspondence; answers RQ6.	A general response orientation and the assurance conditions that connect capability recognition with adoption judgment.
Cross-dimensional comparison	Compares review evidence with the survey's capability–adoption distinction and identifies corresponding assurance conditions.	Research gaps, future directions, and the socio-technical governance framework with design requirements.
Institutional-evidence comparison	Maps explicitly documented initiative functions to the five stages and twelve design requirements; supports RQ7.	Framework-applicability assessment and conservative covered/partially-covered stage assignments.

barriers (Joshi and Choudhury 2022; Cu et al. 2023; Avci and Erzurumlu 2023; Desai et al. 2024; Abualhamayl et al. 2024; Yusof et al. 2023; Abualhamayl et al. 2025; Baum 2021; Sreckovic and Wurdinger 2025; Konashevych 2020; Gupta et al. 2020; Kreppmeier et al. 2023; Swinkels 2023; Serogina and Prokopiev 2025). They introduce the strongest regulatory concerns because fractional tokens may resemble investment contracts or securities, requiring KYC/AML, governance, valuation, and investor-protection mechanisms.

Rental, leasing, and mortgage automation systems extend blockchain use beyond ownership transfer to usage rights and financial workflows (Wang et al. 2022; Junaid et al. 2024; Wang et al. 2024; Borse et al. 2024). This family is less represented than registry and ownership-transfer systems but remains essential for marketplace completeness.

Hybrid regulated marketplace architectures combine property listing, token issuance, escrow, document certification, identity verification, external inspection, regulatory approval, and marketplace coordination (Jeong and Ahn 2021; Sabah 2022; Tan and Nguyen 2022; Exter and Radosavljevic 2023; Oza et al. 2024; Subbarayudu and Hemalatha 2024; Jamshed et al. 2025; Le et al. 2025; Preethi et al. 2026; Huh and Kim 2020; Langaliya and Gohil 2023;

Gutierrez and Xu 2023; Timuçin and Biroğul 2023). These are closest to deployable regulated marketplaces but most demanding to implement, requiring integration among registrars, regulators, banks, inspection bodies, valuation entities, platform operators, and identity providers.

DeFi-oriented investment-token frameworks focus on compliant token standards, SPV structures, investor onboarding, whitelisting, transfer restrictions, and DeFi protocol integration (Kranz et al. 2026; Henker et al. 2023; Zhang et al. 2024b). This emerging family provides more explicit treatment of token standards and secondary-market compliance but increases regulatory exposure.

Table 6 summarizes the families.

5.2 RQ2: Ownership, tokenization, certification, and stakeholder models

Ownership representations comprise full ownership or a title record and transactional property rights, which dominate land-registry studies (Banerjee et al. 2022; Yadav and Kushwaha 2021; Mehta et al. 2023; Zeid et al. 2024; Merchant et al. 2022; Stefanović et al. 2022; Mallya et al. 2025); NFT-based unique ownership (Sharma et al. 2024; Vivekrabinson

Table 6 Architectural families identified in the 63-study corpus

Architectural family	Main characteristics	Representative studies
Registry-centric and deed-management systems	Blockchain as tamper-evident record layer for land records, deeds, ownership history, registry updates, and document verification.	(Banerjee et al. 2022; Yadav and Kushwaha 2021; Ncube et al. 2022; Borse et al. 2022; Mallya et al. 2025; Merchant et al. 2022; Stefanović et al. 2022; Soner et al. 2021; Thakur et al. 2020; Zhang et al. 2024a)
NFT-based ownership systems	Unique properties, certificates, or digital twins represented using NFTs or NFT-like mechanisms.	(Aliti et al. 2023; Rana et al. 2022; Sharma et al. 2024; Vivekrabinson et al. 2023; Jayakumar et al. 2025; Hassan et al. 2024; Mitrea et al. 2025; Arcenegui et al. 2023)
Fractional ownership and tokenized investment systems	Property value or participation rights divided into tokens supporting shared ownership, investment, liquidity, and secondary-market possibilities.	(Abualhamayl et al. 2025; Desai et al. 2024; Joshi and Choudhury 2022; Avci and Erzurumlu 2023; Cu et al. 2023; Baum 2021; Abualhamayl et al. 2024; Kreppmeier et al. 2023; Swinkels 2023; Serogina and Prokopiev 2025)
Rental, leasing, and mortgage automation systems	Smart contracts supporting rental agreements, lease records, mortgage tracking, and conditional obligations.	(Wang et al. 2022; Junaid et al. 2024; Borse et al. 2024; Wang et al. 2024)
Hybrid regulated marketplace architectures	Blockchain combined with escrow, certification, identity verification, inspection, regulatory approval, and marketplace coordination.	(Exter and Radosavljevic 2023; Jamshed et al. 2025; Jeong and Ahn 2021; Tan and Nguyen 2022; Oza et al. 2024; Subbarayudu and Hemalatha 2024; Langaliya and Gohil 2023; Gutierrez and Xu 2023; Timuçin and Biroğul 2023; Huh and Kim 2020; Sabah 2022; Le et al. 2025; Preethi et al. 2026)
DeFi-oriented investment-token frameworks	Real-estate investment tokens for compliant issuance, investor onboarding, transfer restrictions, security-token standards, and DeFi integration.	(Kranz et al. 2026; Henker et al. 2023; Zhang et al. 2024b)

et al. 2023; Jayakumar et al. 2025; Gadagkar and Begum 2024; Mitrea et al. 2025; Hassan et al. 2024); fractional ownership or shared investment (Abualhamayl et al. 2025; Desai et al. 2024; Joshi and Choudhury 2022; Avci and Erzurumlu 2023; Zhang et al. 2024b; Serogina and Prokoviev 2025); rental or lease rights and financing-related claims (Wang et al. 2022; Junaid et al. 2024; Yusof et al. 2023; Arcenegui et al. 2023); and security-token participation rights (Kranz et al. 2026; Henker et al. 2023; Kreppmeier et al. 2023; Swinkels 2023).

Tokenization models differ by ownership meaning and transaction purpose. Registry-centric studies may use a blockchain record without a token. NFT-based studies use non-fungible tokens for unique property assets or certificates. Fractional investment studies use fungible or fractional tokens and hybrid models. Security-token-oriented models, including ERC-20- and EIP-1400-oriented designs, appear when tokenized participation is treated as a regulated investment instrument (Avci and Erzurumlu 2023; Cu et al. 2023; Kranz et al. 2026; Henker et al. 2023). Critically, the token standard does not determine legal meaning — a token may represent title, certification, investment participation, or only a pointer to off-chain documentation.

Certification mechanisms combine on-chain records or hashes with off-chain documents, IPFS or decentralized storage, content identifiers, and oracle-supported verification (Banerjee et al. 2022; Sharma et al. 2023; Gadagkar and Begum 2024; Abualhamayl et al. 2024). Digital signatures, SSI credentials, and authoritative registry or government verification strengthen the connection between technical evidence and recognized property records (Henriques and Correia 2025; Mallya et al. 2025; Serogina and Prokoviev 2025). Regulated investment designs add issuer validation, whitelisting, regulator-approved certification, and prospectus

or SPV validation (Kranz et al. 2026; Henker et al. 2023). The resulting design priority is a governed process for correction, revocation, and legal recognition of anchored records.

Among studies reporting stakeholder roles, structures scale with architectural complexity: registry-centric systems involve buyers, sellers, registrars, land authorities, notaries, and validators; fractional systems add investors, platform operators, issuers, developers, SPVs, valuers, and governance participants; DeFi frameworks add KYC/AML, technology, custody, and compliance providers; and hybrid marketplaces include regulators, financial institutions, inspection and valuation bodies, and dispute-resolution actors (Mallya et al. 2025; Serogina and Prokoviev 2025).

Table 7 consolidates these models. Ownership, tokenization, certification, and stakeholder structures are tightly connected to legal meaning, regulatory exposure, and institutional participation rather than being merely technical choices.

5.3 RQ3: Transaction types

The corpus supports a wide range of transaction types: registration, buying and selling, ownership transfer, registry update, renting and leasing, mortgage or collateral processing, financing and investment, resale or secondary trading, settlement, investor onboarding, and dispute resolution. Registry-centric systems focus on registration and deed management. NFT-based systems enable token-linked transfer of certificates and digital twins, though legal effect depends on off-chain recognition. Fractional systems support investment participation, resale, and secondary-market exchange, requiring valuation, governance, and exit mechanisms. Security-token and DeFi systems add issuance,

Table 7 Synthesis of ownership, tokenization, certification, and stakeholder models

Model dimension	Dominant categories	Interpretation
Ownership model	Full ownership or title record; transactional property rights; NFT-based unique ownership; fractional ownership or shared investment; rental or lease rights; financing-related claims; security-token participation rights.	Ownership meaning is often ambiguous: the token may represent title, certification, investment participation, usage rights, or a regulated security-like interest.
Tokenization model	No token; NFT; fungible or fractional token; hybrid token model; security-token-oriented model; ERC-20/EIP-1400-oriented model.	Token choice depends on whether the system supports certification, unique ownership, fractional investment, regulated participation, or a hybrid rights model.
Certification mechanism	On-chain record or hash; IPFS or decentralized storage; content identifier; oracle-supported verification; digital signatures or SSI credentials; authoritative registry or government verification; regulator-approved certification; prospectus/SPV validation.	Certification combines on-chain evidence with off-chain documents, but legal recognition, correction, revocation, and official validation remain design gaps.
Stakeholder structure	Buyer or seller; investor, tenant, or landlord; registrar, land authority, or notary; regulator; financial institution; inspection or valuation body; platform operator; issuer, developer, or SPV; KYC/AML provider; validator, technology provider, or custody provider.	Advanced tokenization models require broader institutional participation, clearer accountability, and stronger governance.

whitelisting, and DeFi interaction (Kranz et al. 2026; Henker et al. 2023). Rental, leasing, and mortgage systems support usage-right and financial workflows. Hybrid marketplaces combine listing, verification, escrow, inspection, approval, transfer, and audit.

The key finding is that real-estate tokenization is not a single transaction model — its legal, technical, compliance, and governance requirements change fundamentally depending on whether the system supports title transfer, digital certification, economic participation, security-token issuance, usage rights, or marketplace coordination.

5.4 RQ4: Technology models

Ethereum and EVM-compatible platforms recur among implementation-oriented studies because of their smart-contract and token-standard support. ERC-721 appears in NFT-based property representation; ERC-20- and EIP-1400-oriented standards and hybrid models appear in fractional and security-token-oriented systems (Abualhamayl et al. 2025; Desai et al. 2024; Sharma et al. 2024; Oza et al. 2024; Kranz et al. 2026). Permissioned blockchains appear where government or institutional control is required (Borse et al. 2022; Junaid et al. 2024; Henriques and Correia 2025). Recent work broadens the landscape with EVM-compatible deployments, batch transfers, and gasless transactions (Kranz et al. 2026).

The technology taxonomy also captures PoA, Geographic Proof of Authority, and PBFT consensus; SSI, DIDs, and verifiable credentials; and cross-chain or multi-chain integration (Mallya et al. 2025; Henriques and Correia 2025; Serogina and Prokoviev 2025). Post-quantum-ready mechanisms are retained as a forward-looking technology category for property records requiring long-horizon verifiability (Fernández-Caramés and Fraga-Lamas 2020).

Smart contracts support ownership transfer, escrow, rental agreements, mortgage workflows, token issuance, compliance, and conditional execution, though depth varies from conceptual workflows to implemented prototypes. Escrow contracts are particularly important in hybrid marketplaces for conditional fund release (Sabah 2022; Le et al. 2025; Preethi et al. 2026).

IPFS and decentralized storage anchor document hashes on-chain while keeping sensitive legal documents off-chain (Banerjee et al. 2022; Sharma et al. 2023; Gadagkar and Begum 2024; Abualhamayl et al. 2024), though this raises design questions around access, correction, revocation, and legal admissibility. Privacy-enhancing mechanisms such as zero-knowledge proofs and self-sovereign identity appear in a small number of studies (Jeong and Ahn 2021; Henriques and Correia 2025). Long-term security readiness was not reported in any of the 63 coded studies — a significant gap given that real-estate records may need to remain valid for decades (Fernández-Caramés and Fraga-Lamas 2020).

5.5 RQ5: Adoption gaps

The review and cross-dimensional synthesis identify twelve adoption and deployment gaps. *Legal enforceability* is the most critical: many studies assume blockchain records represent property rights without explaining how they become legally binding within registries, courts, or regulated investment markets (Avci and Erzurumlu 2023; Mezquita et al. 2022; Konashevych 2020; Kranz et al. 2026). *Registry integration* is weak: most proposals do not explain synchronization with cadastral systems, title databases, tax systems, and notarial processes. *Identity and KYC/AML*: wallet control does not establish legal identity, beneficial ownership, or investor eligibility (Shuaib et al. 2022; Avci and Erzurumlu 2023; Kranz et al. 2026; Henriques and Correia 2025). *Privacy*: property records contain sensitive identity, ownership, financial, and legal data requiring privacy-preserving certification rather than unrestricted on-chain disclosure. *Valuation and market transparency*: fractional systems require reliable appraisal, disclosure, pricing, and income distribution that most studies do not address (Cu et al. 2023; Yusof et al. 2023; Kreppmeier et al. 2023; Swinkels 2023). *Governance and dispute resolution*: smart contracts cannot handle exceptions, inheritance disputes, failed inspections, or regulatory interventions without institutional escalation procedures. *Scalability, cost, and formal verification*: evidence is concentrated in prototypes (27 studies) and simulation or experimental evaluations (22 studies), with no deployed or operational studies and limited formal verification or smart-contract auditing. *Key management and custody*: systems assume secure private-key management but real-estate transactions involve high-value assets where lost keys or compromised wallets create severe legal consequences. *Interoperability*: tokenized systems must connect with registries, identity providers, payment systems, inspection records, valuation platforms, and DeFi environments. *Public trust and usability*: empirical real-estate blockchain research identifies perceived usefulness, trust, and institutional endorsement as relevant acceptance factors (Yeoh et al. 2024; Yang et al. 2025); technical feasibility alone does not establish adoption. *Post-quantum readiness*: long-term security and cryptographic migration planning were not reported in the coded corpus despite the long lifespan of property records (Fernández-Caramés and Fraga-Lamas 2020). *Dynamic security monitoring*: the structured extraction matrix records explicit security mechanisms or assumptions in 21 studies, while 42 entries use the predefined not reported code. One prototype applies machine-learning-assisted validation to land registration (Polisetti et al. 2025), and another uses dynamic trust scoring, weighted validation, and Byzantine-fault testing (Mallya et al. 2025). These contributions address validation and consensus assurance, while integrated marketplaces introduce a broader

runtime requirement spanning transaction behavior, identity, access, KYC/AML, escrow, oracle, and authoritative-registry events.

Table 8 summarizes the gaps and most affected system families.

Table 9 presents the thematic adoption-barriers matrix across architectural families for four cross-cutting barriers: legal enforceability, scalability and cost, key management and custody, and interoperability.

6 Survey-based respondent perceptions

The survey findings address RQ6 through closed-item distributions, response consistency, qualitative themes, and respondent-level correspondence across 153 responses from Saudi Arabia. The analysis examines perceptions of trust, use, investment, automation, transparency, and token-enabled transactions in a highly regulated real-estate setting.

6.1 Closed-ended findings

Table 10 reproduces the full English translation of each Arabic item and summarizes its Likert-scale results (agreement = responses 4–5; neutral = 3; disagreement = 1–2).

Q1 examines blockchain as an alternative source of trust to central real-estate authorities, while Q2 addresses its capacity to perform a government documentation role. Q3 concerns motivation to use a blockchain application instead of traditional methods. Q6 addresses a single fully automated environment for the principal transaction services, and Q7 addresses whether transparency and traceability would increase respondents' trust. Disagreement therefore reflects the full proposition stated in each item and can coexist with recognition of an individual technical capability.

Agreement remained below 25% and disagreement exceeded 50% across all eight items. The highest relative agreement concerned blockchain-secured investment outside the Kingdom (Q4, 24.8%). The strongest caution concerned the

Table 8 Main adoption gaps identified in the systematic review

Gap	Observed limitation	Most affected families
Legal enforceability	Token–right mappings require explicit official legal recognition.	NFT-based; fractional investment; security-token; hybrid marketplaces.
Registry integration	Synchronization with official registries, cadastral systems, and legacy records remains unspecified.	Registry-centric; NFT-based; hybrid marketplaces.
Identity and KYC/AML	Wallet control requires links to legal identity, beneficial ownership, and transaction eligibility.	Fractional investment; DeFi-oriented; hybrid marketplaces; cross-border models.
Privacy	Sensitive property, identity, financial, and legal data require protected storage and linkage.	Registry-centric; NFT; hybrid marketplaces.
Valuation and market transparency	Fractional tokens require appraisal, disclosure, pricing, and governed income distribution.	Fractional ownership; security-token; DeFi-oriented.
Governance and dispute resolution	Exceptions, conflicts, corrections, and regulatory intervention require institutional procedures beyond contract automation.	Fractional ownership; hybrid marketplaces; DeFi-oriented.
Scalability, cost, and formal verification	Evidence is prototype- and simulation-dominant; operational evaluation, contract audits, and formal verification remain scarce.	All families, especially marketplaces and DeFi-oriented.
Key management and custody	Transaction workflows require explicit custody, key management, and recovery mechanisms.	NFT-based; fractional investment; DeFi-oriented; hybrid marketplaces.
Interoperability	Registries, identity and payment systems, DeFi protocols, and cross-chain environments require coordinated interoperability.	Registry-centric; hybrid marketplaces; DeFi-oriented.
Public trust and usability	Adoption combines perceived usefulness, legal confidence, institutional endorsement, and technical feasibility.	All families.
Post-quantum readiness	Long-lived property records require cryptographic migration planning, which the coded corpus leaves unreported.	Registry-centric; tokenized ownership; hybrid marketplaces.
Dynamic security monitoring	Validation, consensus, integrity, and access controls require continuous monitoring and governed response across marketplace events.	Hybrid marketplaces; fractional and security-token systems; DeFi-oriented frameworks; cross-border models.

Table 9 Thematic adoption-barriers matrix across architectural families

System family	Legal enforceability	Scalability and cost	Key management	Interoperability
Registry-centric	High: on-chain records must align with registries and title law.	Moderate: infrequent updates still require efficient storage and verification.	Moderate: institutional controls for registrar and official keys.	High: cadastral, title, tax, and legal-system integration.
NFT-based ownership	High: token transfer requires legally valid title transfer.	Moderate: minting and transfer costs affect usability.	High: custody loss can create ownership disputes.	High: links to registries, document stores, and legal recognition.
Fractional ownership and investment	High: tokens may represent regulated or security-like interests.	High: transfers and income distribution require efficient execution.	High: investor custody, recovery, and fraud prevention.	Moderate to high: payment, compliance, valuation, and secondary-market integration.
Rental, leasing, and mortgage	Moderate: enforceability depends on recognized contract obligations.	Moderate: recurring payments and updates increase transaction load.	Moderate: secure access and recovery for tenants, landlords, and lenders.	High: payment, mortgage, and legal-documentation integration.
Hybrid regulated marketplaces	High: legally significant steps require validation and dispute handling.	High: multi-step workflows create transaction, storage, and performance demands.	High: role-based custody and recovery across users and institutions.	High: registry, identity, inspection, finance, and dispute integration.
DeFi-oriented investment-token	High: security-token, SPV, eligibility, and transfer rules must be enforceable.	High: DeFi interaction, batch transfer, gas cost, and throughput.	High: reconciliation of self-custody and institutional custody.	High: token-standard, DeFi, compliance, and off-chain legal interoperability.

Table 10 Full survey items and exploratory results from 153 responses (A = agreement; N = neutral; D = disagreement; M = mean)

Item	Full translated statement	Taxonomy linkage	A	N	D	M	SD
Q1	I trust that the technology can serve as an alternative source of trust to a central authority, such as real-estate authorities.	Adoption barrier; stakeholder structure	15.0%	24.8%	60.1%	2.18	1.26
Q2	I believe that blockchain technology is capable of performing the role of government authorities in documenting real-estate ownership securely and reliably.	Certification mechanism; adoption barrier	19.0%	22.9%	58.2%	2.22	1.27
Q3	Having a blockchain-based real-estate application would motivate me to use it instead of traditional methods.	Adoption barrier; marketplace architecture	14.4%	23.5%	62.1%	2.16	1.32
Q4	I would like to invest in real estate outside the Kingdom if the process were secured by blockchain technology.	Transaction type; investment participation	24.8%	22.2%	52.9%	2.42	1.46
Q5	I see a need to open fractional real-estate investment to enable small investors (fractional ownership).	Ownership model; tokenization model	13.7%	20.9%	65.4%	2.07	1.24
Q6	It is important to me that all real-estate services — sale, purchase, documentation, and payment — are available in one place and fully automated.	Hybrid marketplace architecture	13.1%	11.1%	75.8%	1.80	1.21
Q7	I believe that transparency and the ease of tracing real-estate operations through blockchain would increase my trust in the real-estate system.	Certification mechanism; adoption barrier	9.8%	15.7%	74.5%	1.84	1.15
Q8	I believe that using digital tokens — NFTs or fractional tokens — can facilitate real-estate buying and selling.	Tokenization model; transaction type	16.3%	24.2%	59.5%	2.19	1.31

unified and fully automated environment in Q6 (75.8% disagreement) and the trust effect stated in Q7 (74.5% disagreement). The aggregate pattern indicates caution toward institutional substitution, complete automation, and technology-generated trust while preserving recognition of particular blockchain capabilities.

The eight items demonstrated high aggregate response consistency. Cronbach's alpha was $\alpha = 0.933$, with a 95% bootstrap confidence interval of [0.910, 0.949] based on 2,000 resamples; standardized alpha was 0.935. Corrected item-total correlations ranged from 0.708 to 0.824, and alpha-if-item-deleted ranged from 0.920 to 0.930. Each item therefore contributed to the coherent orientation. Because the statements address related but distinct propositions, alpha is interpreted as aggregate response consistency; construct-specific scale validation remains a confirmatory research direction.

An unrotated single-factor assessment using principal-component extraction found that the first component explained 68.7% of total item variance. The eight loadings were positive and ranged from 0.773 to 0.871. This dominant component indicates a substantial combined signal from the general caution-to-favorability orientation and the shared question-

naire method. Subsequent analyses therefore interpret the coefficients within this general response disposition.

Table 11 reports the complete-sample Spearman correlations. All 28 associations were positive ($\rho = 0.454$ – 0.781 ; Holm-adjusted $p < 0.001$). Together with the reliability and single-factor results, this pattern shows that respondents who were relatively more favorable toward one proposition also tended to evaluate the others more favorably. The coef-

Table 11 Spearman correlations among survey items ($n = 153$)

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Q1	1.000							
Q2	0.741	1.000						
Q3	0.698	0.733	1.000					
Q4	0.645	0.583	0.637	1.000				
Q5	0.678	0.734	0.710	0.595	1.000			
Q6	0.542	0.642	0.593	0.454	0.680	1.000		
Q7	0.585	0.648	0.701	0.561	0.703	0.781	1.000	
Q8	0.636	0.663	0.672	0.696	0.630	0.598	0.653	1.000

All correlations are positive. Holm-adjusted $p < 0.001$ for all 28 pairwise tests

ficients therefore express a coherent overall orientation, with construct-level relationships treated as exploratory patterns.

Several coefficients provide informative exploratory comparisons. Q1 and Q2 were closely associated ($\rho = 0.741$), while token usefulness (Q8) was associated with cross-border investment interest (Q4; $\rho = 0.696$). The strongest association, between Q6 and Q7 ($\rho = 0.781$), shows similar ranking of two system-level propositions. The dominant general factor guides these comparisons toward hypothesis development, with directional relationships reserved for validated construct-specific evaluation.

All 153 responses were retained because a uniform profile can represent a genuine position. Fifty-three respondents (34.6%) selected one category throughout, including 42 who selected 1 for all items. A diagnostic sensitivity assessment examined their influence while preserving the complete sample as the primary analysis. Among the 100 differentiated profiles, alpha remained 0.884, the first component explained 55.8% of variance, and the Spearman correlations remained positive ($\rho = 0.234$ – 0.702). The coherent orientation therefore persisted beyond the invariant profiles, with their contribution visible in the higher complete-sample coefficients.

6.2 Open-ended perceived advantages and concerns

The two open-ended items requested the most important additional advantage and concern, explicitly directing respondents toward a consideration “not mentioned previously.” They therefore extend the closed items. An absent concept was coded as unreported, while each explicitly stated concept was preserved; answers containing more than one stated idea retained each idea (Tables 12 and 13).

The qualitative findings show simultaneous recognition of technical value and attention to deployment conditions. Security-related concerns support continuous monitoring

and accountable response workflows. Regulatory and usability concerns support authoritative integration, clear interfaces, and governed correction. Together, the themes connect respondent perceptions with the legal, operational, and user-protection gaps identified in Section 5.

6.3 Correspondence between closed and open responses

A respondent-level correspondence analysis identified open advantages that explicitly mentioned a concept related to each closed item and compared them with the same respondent’s closed response. The comparison remains descriptive because the open prompts requested additional considerations. It distinguishes recognition of a capability from agreement with the complete proposition in the corresponding item.

For Q1, eight respondents identified a trust-related capability; seven disagreed with the complete authority-alternative proposition and one was neutral. Their accompanying comments favored institutional connection and protection alongside technical trust mechanisms. For Q2, 12 respondents mentioned secure documentation or official linkage: three agreed, three were neutral, and six disagreed. This mixed pattern distinguishes reliable digital records from assigning blockchain a governmental documentation role.

For Q3, 21 respondents identified ease, speed, access, or reduced procedure. Fifteen disagreed with replacing traditional methods, five were neutral, and one agreed. Operational benefits were therefore recognized while adoption remained connected to system readiness and user assurance. For Q4, the two cross-border mentions accompanied disagreement with personal investment willingness. For Q5, the two fractional-ownership mentions accompanied disagreement with the stated need to open this model to small investors. These small groups show how market-level

Table 12 Main themes from the open-ended responses

Theme	Count	Interpretation
Security, trust, and reliable documentation (advantage)	22	Secure documentation and stronger trust under appropriate governance.
Speed, ease, and process efficiency (advantage)	18	Reduced process friction and improved coordination.
Transparency, traceability, and ownership history (advantage)	16	Visible ownership history and traceable records.
Integration, unification, or explicit automation (advantage)	7	Unified services, coordinated procedures, or automated execution.
Price visibility and reduced manipulation (advantage)	13	Clearer pricing and reduced intermediary influence.
Cybersecurity, hacking, fraud (concern)	13	Security exposure across platforms, accounts, and transactions.
Weak regulation, legal control (concern)	13	Official approval and regulatory governance as trust conditions.
Limited understanding or usability (concern)	14	Clear concepts, interfaces, and user support as adoption conditions.
Privacy and data exposure (concern)	11	Protection of sensitive property, financial, and identity data.
Trust, reliability, technology readiness (concern)	10	Reliable operation and ecosystem readiness.
Difficulty correcting errors or recovering rights (concern)	6	Legally valid correction and rights-recovery procedures.

Table 13 Correspondence between explicit open-response components and closed-item positions

Q	Explicit component in an open advantage	<i>n</i> (% of 153)	Agree	Neutral	Disagree
Q1	Trust, reliability, or non-intermediated verification	8 (5.2%)	0 (0.0%)	1 (12.5%)	7 (87.5%)
Q2	Secure ownership documentation or official linkage	12 (7.8%)	3 (25.0%)	3 (25.0%)	6 (50.0%)
Q3	Ease, speed, access, or reduced traditional procedure	21 (13.7%)	1 (4.8%)	5 (23.8%)	15 (71.4%)
Q4	Cross-border or global property access	2 (1.3%)	0 (0.0%)	0 (0.0%)	2 (100%)
Q5	Fractional or divided ownership	2 (1.3%)	0 (0.0%)	0 (0.0%)	2 (100%)
Q6	Integration, unification, or explicit automation	7 (4.6%)	2 (28.6%)	1 (14.3%)	4 (57.1%)
Q7	Transparency, traceability, or ownership history	16 (10.5%)	3 (18.8%)	0 (0.0%)	13 (81.3%)
Q8	Tokenized or fractional digital ownership	3 (2.0%)	1 (33.3%)	1 (33.3%)	1 (33.3%)

Percentages in the final three columns use the open-mention count in the corresponding row as the denominator. The small Q4, Q5, and Q8 groups support case-level interpretation

recognition can remain distinct from personal or policy endorsement.

For Q6, seven respondents mentioned integration, unification, or automation: two agreed, one was neutral, and four disagreed. The four disagreeing respondents also raised a practical condition concerning system integration, cybersecurity, service continuity, or legal governance. Their responses support coordinated services through staged, assured automation.

For Q7, 16 respondents mentioned transparency, traceability, or ownership history. Three agreed and 13 disagreed that these capabilities would increase their trust; 12 of the 13 also stated a substantive concern. These concerns centered on legal protection, technical security, or operational assurance. The pattern values transparent and traceable records while locating trust within a wider governance environment.

For Q8, the three tokenization-related mentions were distributed evenly across agreement, neutrality, and disagreement. Their comments connect transactional usefulness with accurate registration, rights recovery, and appropriate asset selection. The group supports descriptive interpretation at the case level.

Across Q1–Q8, the correspondence analysis shows that recognition of a related technical capability can coexist with a different judgment about the complete institutional, behavioral, or market proposition. The open responses contextualize the closed distributions by identifying the assurance conditions that accompany adoption judgments.

6.4 RQ6: Respondent perceptions of blockchain-based real-estate tokenization

The survey answers RQ6 by identifying a predominantly cautious and internally coherent orientation toward blockchain-based real-estate tokenization in the observed sample. The strongest relative interest concerned blockchain-secured cross-border investment, while the greatest caution con-

cerned a fully automated service environment and the stated effect of transparency and traceability on trust.

Reliability and single-factor findings show that respondents largely varied along a general cautious-to-favorable orientation. The positive Spearman coefficients capture coherent movement across the items, and the sensitivity assessment shows that this orientation persists among differentiated response profiles.

The open–closed comparison supplies the main interpretive insight. Respondents recognized particular technical and market benefits while linking adoption to legal authority, operational assurance, and usable rights protection. Technical capability and adoption judgment therefore operate at different levels. This finding supports an institution-connected, staged pathway in which functionality advances with legally recognized documentation, regulatory accountability, security assurance, and understandable user services.

7 Discussion: Cross-dimensional synthesis

The combined findings address the integration gap by showing that architecture, represented rights, transactions, tokenization, certification, stakeholders, technology, and adoption are interdependent. Registry systems require trusted certification and official integration; fractional platforms add investor eligibility, valuation, governance, and approval; and DeFi frameworks further need compliance-aware standards, transfer restrictions, and interoperability. The synthesis thus explains how technical, legal, institutional, market, and perception conditions jointly shape regulated tokenization.

The 63-study corpus shows a clear evolution from land-record digitization toward marketplace-oriented tokenization. Early studies focused on registry integrity, deed management, and tamper-evident records (Banerjee et al. 2022; Yadav and Kushwaha 2021; Thakur et al. 2020). More recent studies extend toward NFT-based representation, frac-

tional investment, security-token models, rental and mortgage automation, DeFi frameworks, and hybrid marketplace workflows (Abualhamayl et al. 2025; Desai et al. 2024; Junaid et al. 2024; Oza et al. 2024; Kranz et al. 2026; Serogina and Prokoviev 2025). This evolution changes the nature of the design problem: blockchain transitions from a record-keeping infrastructure to a transaction coordination environment in which property rights, financial participation, compliance, and institutional accountability must be jointly managed.

Ownership model and transaction type are especially interdependent. Full-ownership models align with registry updates and title transfer. NFT-based models suit unique property representation but require legal interpretation before supporting actual ownership transfer. Fractional models align with investment participation but create regulatory obligations because tokens may resemble securities (Avci and Erzurumlu 2023; Cu et al. 2023). Security-token and DeFi models further require SPV structures, investor onboarding, and compliance-aware token standards (Kranz et al. 2026; Henker et al. 2023). A single tokenization model cannot serve all real-estate transactions — design must be selected by transaction purpose and the legal nature of the right represented.

Certification is a critical design gap. Hash anchoring and IPFS-based designs improve auditability (Banerjee et al. 2022; Sharma et al. 2023; Gadagkar and Begum 2024), but they do not solve legal recognition, correction, revocation, or dispute handling. Real-estate records require updates for court decisions, inheritance, administrative correction, and fraud investigation. This reveals a distinctive design requirement for real estate: technical immutability must coexist with legal revisability — a constraint absent in ordinary digital asset systems.

Technology choices create governance consequences. Public blockchains raise privacy, compliance, cost, and key-management concerns. Permissioned architectures may be more suitable for regulated real-estate contexts because they allow official actors to control validation and governance (Borse et al. 2022; Junaid et al. 2024). Smart contracts automate escrow, payments, and compliance checks but cannot resolve disputed off-chain facts, valuation disagreements, or regulatory exceptions. This supports treating tokenization as market-infrastructure augmentation rather than institutional replacement — consistent with policy analysis showing tokenization may reduce intermediary-process costs rather than eliminate intermediaries (Agur et al. 2025).

Scalability and cost extend beyond transaction fees to infrastructure, secure storage, registry and identity integration, KYC/AML, smart-contract assurance, security monitoring, custody, user support, training, maintenance, and cryptographic migration. Evaluation should relate throughput, latency, finality, availability, and storage growth to

complete-workflow cost and service requirements, then compare processing time, transparency, errors, user experience, and regulatory assurance with existing processes.

Security evidence distinguishes pre-deployment from runtime assurance. Twenty-one studies record mechanisms or assumptions, including ML-assisted land-registration validation (Polisetti et al. 2025) and dynamic validator trust scoring (Mallya et al. 2025), strengthening validation, consensus, access, or integrity. Stage 4 additionally requires continuous correlation of transaction, identity, access, KYC/AML, escrow, oracle, and registry events. Layered runtime assurance can combine deterministic rules with AI/ML anomaly detection for unusual transfers, identity–wallet inconsistencies, access failures, abnormal investment flows, and on-/off-chain divergence. Auditable, role-authorized workflows should govern alerts, investigation, protective controls, and legally significant action.

The survey complements the review by showing how technical capabilities and adoption judgments interact in a regulated market. Agreement remained below 25% and disagreement exceeded 50% across the eight items. High reliability and a dominant first component indicate a coherent general orientation, so the positive Spearman coefficients are interpreted primarily as relative movement from caution to favorability.

The respondent-level comparison supplies the more specific insight. Participants could value a capability, such as traceable records or coordinated services, while reserving judgment on authority substitution, complete automation, or personal trust. Their open concerns locate adoption within institutional assurance, reliable operation, and user protection. Together, these findings support authoritative integration, staged automation, continuous security assurance, governed correction, and clear user services.

These interpretations characterize the aggregate sample. Future stratified or probability-based replication can add demographic and professional variables, subgroup comparisons, and multivariable analysis of how technical familiarity and market role shape adoption.

Evidence across all six families is concentrated at prototype and simulation/experimental levels, positioning operational deployment as a subsequent validation stage. Legally and technically demanding fractional, hybrid, and DeFi models require governance, identity, valuation, investor protection, custody, disputes, interoperability, and smart-contract assurance. The matrix also reveals limited long-term security readiness across the corpus despite the decades-long authority of property records. Public-key signatures, credentials, wallet approvals, and off-chain key establishment require standardized post-quantum migration. Document anchors require hash and commitment agility through algorithm identifiers, versioning, adequate strength, and append-only re-anchoring as guidance changes. Authorized re-attestation

linking the original and replacement anchors to the authoritative document preserves the audit trail, confidentiality, integrity, and evidentiary continuity (Fernández-Caramés and Fraga-Lamas 2020; National Institute of Standards and Technology 2024b, a; Barker et al. 2026).

Official institutional evidence complements this academic maturity profile by showing how the taxonomy's technical and governance patterns are being instantiated in supervised pilots and services. Sweden's Lantmäteriet pilot and Georgia's National Agency of Public Registry illustrate registry-linked certification, cryptographic anchoring, and the importance of pre-existing digital records and digital identity (World Bank 2020). The United Kingdom's HM Land Registry Digital Street prototype connected identity verification, smart-contract coordination, payment, conveyancing actors, and registry updating in an end-to-end property-transfer demonstration (Tombs 2019). Singapore's EDB-profiled InterOpera initiative illustrates regulated fractional-investment functionality, investor verification, compliance-aware smart contracts, and digital custody (Lee and Chin 2022). Dubai's Prypc Mint initiative combines licensed tokenized investment with regulatory coordination and a controlled secondary-market phase (Dubai Land Department 2025; Virtual Assets Regulatory Authority 2026; Government of Dubai Media Office 2026). Saudi Arabia has completed an initial title-deed tokenization and announced a broader national architecture linking the authoritative registry, fractional ownership, payment verification, smart contracts, open interfaces, and regulated digital market participation (Saudi Press Agency 2025a, b).

These disclosed features align with the peer-reviewed taxonomy, indicating that it provides a research foundation for institutional implementation, while initiatives contribute operational coordination, authorization, registry linkage, and controlled deployment. Qatar and Abu Dhabi further illustrate supervised tokenization, fractional participation, exchange functions, registration, title validation, and digital-identity integration (Qatar Financial Centre 2026; Abu Dhabi Real Estate Centre 2026). India's Property Chain and Colombia's land-allocation certificate pilot reinforce registry-linked adoption through transaction histories and verifiable public records (National Informatics Centre 2026; Ministry of Information and Communications Technologies of Colombia 2022). Their emphasis is record integrity and registry coordination, complementing investment tokenization in Singapore, Dubai, Qatar, Abu Dhabi, and Saudi Arabia. The next evaluation priorities are longitudinal performance, runtime security, custody and recovery, formal assurance, legal effectiveness, and user outcomes.

Relative to the research streams in Table 1, the findings confirm the breadth and uneven maturity identified by broad blockchain–real-estate reviews while explicitly connecting architecture, rights, transactions, certification, stakeholders,

and adoption (Saari et al. 2022a, b). Land-administration reviews emphasize registry integrity, institutional authority, and regulatory readiness (Mezquita et al. 2022; Ansah et al. 2023; Riaz and Ikram 2023); this synthesis treats them as foundations and adds token-right definition, valuation, investor eligibility, custody, secondary transfer, and disputes. The TOE-based review of Jaouhari et al. (2025) organizes applications, drivers, and barriers; the present taxonomy links them to architectural families, maturity, and governance. Broader RWA systematizations emphasize life-cycles, legal interoperability, and economic value (Luo et al. 2026; Chemla and Tinn 2026); this study specifies property-sector needs for authoritative registries, legally revisable certification, inspection, valuation, and institutional dispute resolution.

Prior empirical studies directly operationalize validated technology-adoption models (Yeoh et al. 2024; Yang et al. 2025). The present survey adds exploratory evidence from a highly regulated real-estate setting. Its central contribution is the distinction between recognizing a technical capability and endorsing its broader adoption consequence. This distinction complements established adoption research by emphasizing legal recognition, operational assurance, and visible institutional accountability.

Practically, regulators and land authorities can require defined token rights, authoritative registry integration, correction, and dispute resolution. Platform operators should align architecture and tokens with transaction purpose and embed identity, KYC/AML, privacy, custody and recovery, and compliance-aware contracts. Investment actors should add valuation, SPV or prospectus validation, investor eligibility, and secondary-transfer governance. Bounded institutional pilots can measure legal, technical, operational, and user outcomes before expansion.

The synthesis therefore aligns legally meaningful tokens, transaction-specific workflows, trusted certification, verified identity, privacy-preserving data, institutional governance, custody and recovery, and user-centered design. These conditions ground the gaps in Section 8 and framework in Section 9.

8 Research gaps and future directions

The systematic review findings (RQ5), exploratory survey evidence (RQ6), and cross-dimensional synthesis (Section 7) yield twelve research gaps and corresponding future directions. Together, these gaps form the evidential bridge between the taxonomy and the socio-technical governance framework in Section 9, thereby addressing the evidence-to-design translation gap identified in the related-work analysis.

Gap 1 — Token-right mapping Many studies propose NFTs, fungible tokens, fractional NFTs, or security-token models

without specifying whether the token represents legal title, a digital certificate, a fractional investment claim, a lease right, or a pointer to an off-chain record. Technical token transfer does not automatically produce legally enforceable property transfer (Avci and Erzurumlu 2023; Mezquita et al. 2022; Konashevych 2020). Future research should develop token-right mapping models distinguishing title, certification, investment, lease-right, security, and governance tokens, and examine how each category links to enforceable legal instruments and official approval workflows.

Gap 2 — Registry integration and interoperability with legacy systems Registry-centric studies show blockchain can strengthen record integrity and auditability (Banerjee et al. 2022; Yadav and Kushwaha 2021; Mallya et al. 2025), but most proposals do not explain how on-chain records interact with cadastral databases, title registries, notarial procedures, tax systems, or court decisions. Future research should investigate registry-interoperability architectures covering synchronization mechanisms, official APIs, authority hierarchies, legal-effect rules, correction procedures, and conflict-resolution models when on-chain and off-chain records diverge.

Gap 3 — Identity, KYC/AML, and transaction eligibility. Wallet control does not establish legal identity, beneficial ownership, investor eligibility, or authority to sell (Shuaib et al. 2022). Fractional investment, cross-border participation, security-token issuance, and DeFi interaction amplify money-laundering, fraud, and unauthorized transaction risks (Kranz et al. 2026; Henriques and Correia 2025). Future research should examine how digital identity, verifiable credentials, KYC/AML procedures, role-based authorization, and privacy-preserving proofs can be integrated without exposing unnecessary personal data.

Gap 4 — Privacy-preserving certification Blockchain transparency is presented as a trust-enhancing feature, yet real-estate transactions involve sensitive identity, ownership, valuation, financing, and legal data. Studies using IPFS, document hashes, and zero-knowledge proofs show verification can be separated from full disclosure (Banerjee et al. 2022; Jeong and Ahn 2021; Gadagkar and Begum 2024), but mature privacy models for regulated tokenization remain absent. Future research should develop privacy-preserving certification architectures combining encrypted off-chain storage, selective disclosure, access control, audit trails, and legally valid correction and revocation mechanisms.

Gap 5 — Valuation and market transparency Fractional and tokenized investment systems are justified by liquidity and investor-access benefits (Abualhamayl et al. 2025; Desai et al. 2024; Joshi and Choudhury 2022; Kranz et al.

2026), but most studies do not address valuation governance, transparent pricing, income-distribution rules, or investor-risk disclosure. Without these mechanisms, tokenized markets risk speculation, mispricing, and information asymmetry. Future research should examine independent appraisal, periodic revaluation, secondary-market pricing, revenue transparency, and audit mechanisms for fractional and security-token models.

Gap 6 — Governance and dispute resolution Smart contracts automate predefined rules but cannot handle exceptional cases — failed inspections, financing delays, inheritance disputes, erroneous records, regulatory intervention, or court decisions. Fractional ownership further requires governance for voting, income distribution, exit rights, and conflict management (Cu et al. 2023; Abualhamayl et al. 2024). Future research should develop governance models defining authority, accountability, dispute escalation, correction procedures, emergency intervention, and the relationship between platform governance and official legal authority.

Gap 7 — Scalability, lifecycle cost, evaluation maturity, and formal verification. Evidence is concentrated in 27 prototype studies and 22 simulation or experimental evaluations, making operational deployment validation a priority for future research. High-value workflows involving escrow, ownership transfer, fractional distribution, investor onboarding, registry synchronization, and regulatory approval require assurance under realistic transaction volumes and multi-institutional operating conditions.

Future studies should report end-to-end throughput, transaction and approval latency, settlement finality, peak-load behavior, storage growth, availability, failure recovery, and interoperability latency. Economic evaluation should apply a lifecycle cost model covering on-chain execution, infrastructure, secure storage, registry and identity integration, KYC/AML and compliance operations, auditing, security monitoring, custody and recovery, user training, maintenance, and cryptographic migration. Public, permissioned, and hybrid architectures should be compared with existing registry-workflow baselines using equivalent workload and service requirements. Smart-contract auditing, threat modeling, and formal verification should accompany these performance and cost evaluations for legally significant workflows.

Gap 8 — Key management, custody, and recovery Several systems assume users can securely manage private keys, but real-estate assets are high-value and long-lived. Lost keys, compromised wallets, phishing, or insider misuse create severe legal and financial consequences, particularly in NFT-based, fractional investment, and DeFi-oriented models where custody governs both asset control and transaction

authorization. Future research should investigate institutional custody, multi-signature authorization, recovery procedures, role-based key management, and regulated custodian models.

Gap 9 — Cross-system interoperability Tokenized real-estate systems must interoperate with registries, identity providers, payment systems, valuation platforms, inspection authorities, legal databases, tax systems, banking systems, and DeFi protocols, yet most studies propose isolated platforms without explaining these connections. Future research should examine interoperability architectures, API governance, data standards, oracle governance, cross-platform certification, and mechanisms maintaining consistency between on-chain and off-chain systems.

Gap 10 — Adoption, usability, and public trust Agreement remained below 25% and disagreement exceeded 50% across the eight perception items. High reliability and the dominant first factor identify a coherent general response orientation, while the open–closed correspondence shows that recognition of a capability can coexist with conditional adoption. Future studies can separate these effects through validated multi-item measures grounded in TAM or UTAUT (Davis 1989; Venkatesh et al. 2003), together with stratified and longitudinal designs that examine stakeholder-specific patterns.

Gap 11 — Post-quantum readiness and long-term cryptographic resilience The multi-decade lifetime of property records creates a requirement for sustained confidentiality, authenticity, integrity, and verifiability across changes in cryptographic standards. The coded corpus reveals an evidence gap in long-term security readiness and cryptographic migration planning (Fernández-Caramés and Fraga-Lamas 2020).

Future research should develop and evaluate lifecycle models that combine cryptographic inventories, algorithm identifiers, migration triggers, Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM) key establishment, Module-Lattice-Based Digital Signature Algorithm (ML-DSA) signatures, hybrid transition modes, key rotation, commitment versioning, randomized commitments with controlled opening values, and append-only re-anchoring and re-attestation. Evaluations should measure verification continuity, privacy protection, migration and re-anchoring cost, storage overhead, interoperability, and evidentiary traceability (National Institute of Standards and Technology 2024b, a; Barker et al. 2026).

Gap 12 — Dynamic security monitoring and AI/ML-assisted anomaly detection The structured extraction matrix records explicit security mechanisms or assumptions in 21 studies.

Two particularly relevant examples apply machine-learning-assisted validation to blockchain land registration (Polisetti et al. 2025) and dynamic trust scoring with weighted validation and Byzantine-fault testing (Mallya et al. 2025). These approaches strengthen validation and consensus assurance. An integrated Stage 4 marketplace introduces a broader operational security environment in which transaction, identity, access, KYC/AML, escrow, oracle, and registry events must be correlated continuously.

Continuous analysis of irregular temporal transaction patterns can support the detection of criminal blockchain addresses, as demonstrated for USDT activity on the TRON network (Yadulla et al. 2023). For regulated real-estate tokenization, this temporal-detection principle provides a transferable starting point that requires domain-specific datasets, training, evaluation, and governance.

Future research should compare deterministic rule-based monitoring, statistical baselines, and AI/ML-assisted temporal or graph-based models for detecting fraud, identity spoofing, unauthorized access, and evolving transaction behavior. Evaluations should report precision, recall, F1-score, false-positive rate, detection latency, model drift, adversarial robustness, explainability, privacy impact, auditability, and operating cost. Model outputs should support authorized human and regulatory review through traceable alert, investigation, escalation, and response procedures.

These twelve gaps define the legal, institutional, technical, operational-security, and adoption conditions required for broad regulated deployment. They provide the direct basis for the socio-technical governance framework and twelve design requirements proposed in Section 9.

9 Socio-technical governance framework for regulated real-estate tokenization

The twelve gaps identified in Section 8, together with the exploratory survey distinction between capability recognition and adoption judgment, converge on a single conclusion: blockchain-based real-estate tokenization requires a socio-technical framework that aligns legal authority, institutional accountability, technical infrastructure, market processes, and user trust. This framework is the paper's primary design-oriented contribution. It translates the systematic-review evidence, exploratory perception findings, and cross-dimensional gap analysis into the adoption pathway and design requirements specified by RQ7. The framework is presented as a review-derived design artifact for subsequent expert, case-study, and implementation evaluation. The official-initiative comparison in Section 9.4 provides an additional applicability assessment by mapping disclosed institutional functions to the framework stages and design requirements.

9.1 Framework foundations and central principle

The framework rests on a single design principle: *blockchain should support official authorities rather than replace them*. Real-estate rights are legally recognized through jurisdiction-specific institutions — land registries, courts, notarial bodies, regulators, and financial authorities. A tokenized real-estate marketplace therefore cannot be treated as a decentralized trading environment. It must be designed as a regulated system in which blockchain provides traceability, auditability, certification, and workflow coordination while legal authority remains anchored in officially recognized institutions.

This principle follows from three converging evidence sources. The systematic review identifies recurring requirements for legal enforceability, registry integration, identity, valuation, and governance. Adoption research confirms that institutional support, legal recognition, and public trust govern acceptance in high-value domains more than technical capability (Ansah et al. 2023; Mezquita et al. 2022; Avci and Erzurumlu 2023; Yeoh et al. 2024). Survey respondents were cautious toward authority substitution and complete automation, while their open answers emphasized legal recognition, institutional accountability, and operational assurance. This is consistent with policy analysis of tokenization, which shows that blockchain reduces intermediary-process costs rather than eliminating institutional intermediaries (Agur

et al. 2025). The framework operationalizes this design principle through a five-stage adoption pathway and twelve review-derived design requirements.

The perception evidence grounding the framework was collected in Saudi Arabia, a market characterized by centralized official registration and a single national real-estate regulator. The framework's five-stage pathway and twelve design requirements are therefore most directly grounded in this type of highly regulated, government-centric market context. The design requirements distinguish two classes: jurisdiction-invariant requirements, which address fundamental conditions for any regulated real-estate-tokenization marketplace regardless of legal tradition or institutional setup (legal enforceability, KYC/AML, privacy, scalability, cost, formal verification, dynamic security monitoring, key management, and post-quantum readiness); and context-sensitive requirements, whose specific implementation depends on the local registry architecture and governance model. Jurisdictions with different land-title systems or regulatory cultures can apply the jurisdiction-invariant requirements as minimum conditions and adapt the context-sensitive requirements to their institutional environment.

To make the derivation of the framework explicit, Table 14 summarizes how the main design-requirement groups were derived from the systematic review gaps and the exploratory survey findings.

Table 14 Traceability from evidence to design-requirement groups

Req. group	SLR evidence / gap	Survey-informed design implication
Legal and registry alignment	Token-right ambiguity, weak legal enforceability, limited registry integration, and unclear correction or revocation procedures.	Keep official registries legally authoritative; define token-right meaning before issuance; use blockchain as a certification, audit, and coordination layer.
Identity, compliance, and privacy	Wallet ownership does not prove legal identity, beneficial ownership, transaction eligibility, or KYC/AML compliance; sensitive property data require protection.	Integrate digital identity, KYC/AML, role-based authorization, selective disclosure, off-chain storage, and privacy-preserving verification.
Investment, valuation, and governance	Fractional and security-token models require appraisal, disclosure, investor protection, income distribution, and dispute-resolution mechanisms.	Require regulated valuation, transparent disclosure, investor-governance rules, institutional escalation, and dispute-resolution procedures.
Technical assurance, runtime security, and interoperability	The corpus remains prototype-dominant. Reported security mechanisms primarily address validation, consensus, integrity, or access assurance, while integrated marketplaces require continuous event monitoring alongside benchmarking, auditing, custody, interoperability, and long-term cryptographic planning.	Require performance testing, smart-contract audits, continuous event monitoring, AI/ML-assisted anomaly detection, explainable and auditable alerting, authorized response escalation, custody and recovery mechanisms, registry/payment/API interoperability, crypto-agility, versioned commitments, and governed re-anchoring and re-attestation.
User trust and phased adoption	Respondents showed a coherent cautious orientation while recognizing particular capabilities; open answers linked adoption to institutional, operational, and user assurance.	Provide clear explanations and official validation; stage integration and automation as assurance matures.

9.2 The five-stage adoption pathway

The framework follows a staged logic in which each stage resolves a specific subset of the twelve gaps identified in Section 8 before advancing to the next. Figure 3 illustrates the full pathway.

Stage 1 — Trusted digital certification Blockchain records tamper-evident evidence of property documents, ownership events, inspection records, valuation reports, transaction milestones, and approval events. The official registry remains the legally authoritative title source; blockchain strengthens auditability and reduces document manipulation. This stage directly addresses gaps 4 (privacy-preserving certification) and 2 (registry integration) at low institutional risk, and creates the institutional learning phase needed before blockchain is extended to legally sensitive ownership transfer. It is the minimum viable entry point for any jurisdiction.

Stage 2 — Regulator-gated transaction workflows Blockchain coordinates transaction steps — identity verification, KYC/AML checks, escrow, inspection approval, valuation confirmation, payment confirmation, registrar validation,

and final transfer authorization — through smart contracts whose execution remains conditional on approval from the relevant official institution. This stage addresses gaps 3 (identity and KYC/AML), 6 (governance and dispute resolution), and 7 (scalability and formal verification). It moves blockchain from passive certification to active workflow coordination while preserving legal oversight. This sequencing is consistent with respondents' distinction between valuing coordinated services and accepting complete automation.

Stage 3 — Controlled tokenization Fractional investment, tokenized participation in approved projects, digital certificates, lease rights, and security-token participation are introduced under explicit regulatory supervision. The legal meaning of each token must be defined before issuance — a certificate, investment claim, fractional economic interest, lease right, financing claim, or legally recognized ownership interest. These categories must not be conflated. This stage addresses gaps 1 (token-right mapping), 5 (valuation and market transparency), and 8 (key management and custody), and requires investor rights, governance procedures, valuation mechanisms, and regulatory classification to be

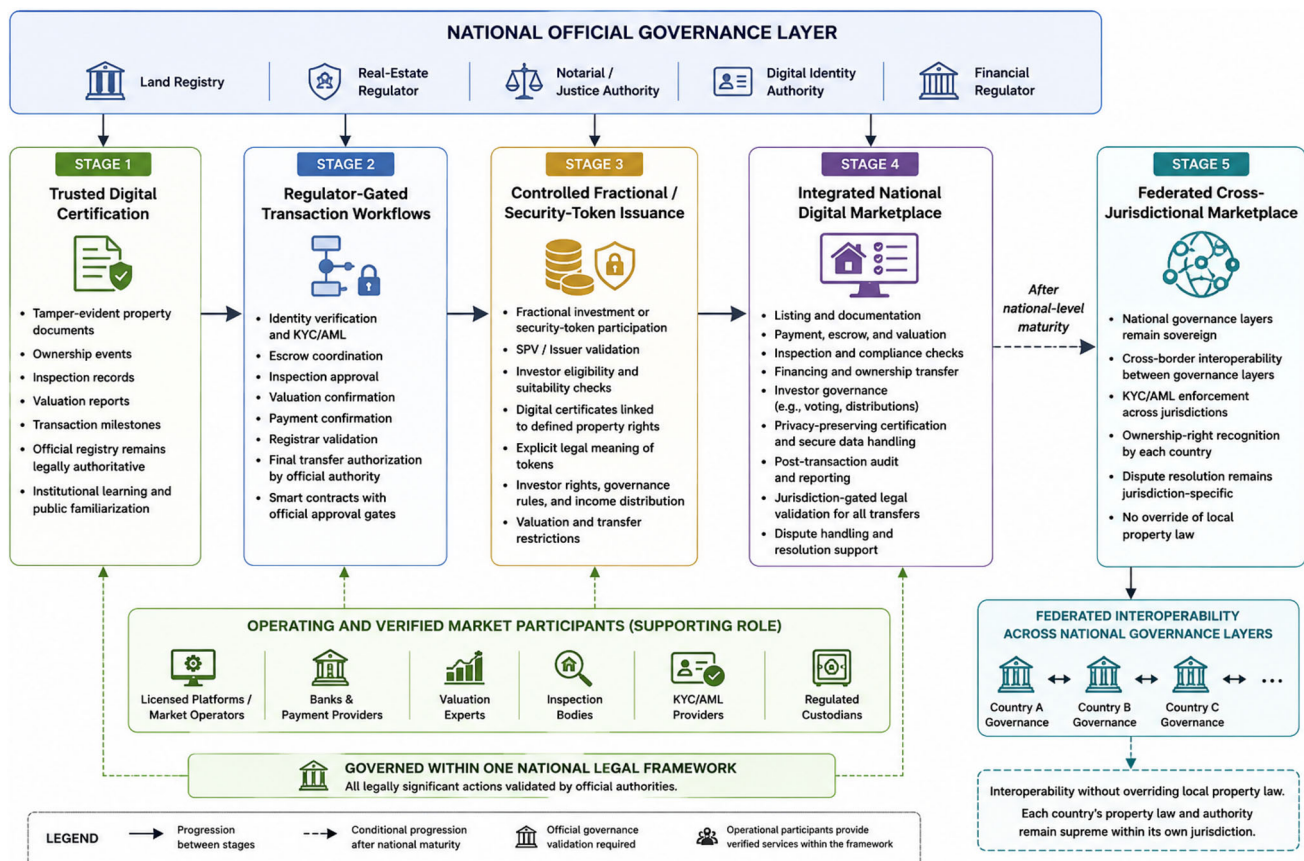


Fig. 3 A jurisdiction-gated and consortium-governed adoption pathway for blockchain-based real-estate tokenization

established before any token is issued (Avci and Erzurumlu 2023; Cu et al. 2023; Kranz et al. 2026).

Stage 4 — Integrated national digital marketplace Listing, documentation, identity verification, payment, inspection, valuation, financing, ownership transfer, fractional governance, dispute handling, and post-transaction audit are coordinated through a unified platform. The marketplace remains jurisdiction-gated: all legally significant transactions are validated under the national legal framework. Blockchain functions as shared coordination and evidence infrastructure; legal validity continues to depend on recognized authorities. This stage addresses gaps 9 (cross-system interoperability), 10 (adoption and usability), and 12 (dynamic security monitoring), together with the full registry-integration requirement, because institutional and market actors interact within one governed environment.

Stage 4 includes a governed security-operations capability that correlates on-chain transactions with identity, access, KYC/AML, escrow, oracle, and authoritative-registry events. Deterministic rules and AI/ML-assisted anomaly detection generate explainable and traceable alerts for authorized review, investigation, regulatory escalation, and proportionate protective action.

Stage 5 — Federated cross-jurisdictional marketplace After national-level maturity is achieved, the model expands to federated cross-border operation. Each participating jurisdiction retains its own official governance layer responsible for property record validation, transfer approval, KYC/AML enforcement, ownership recognition, investment-token supervision, and dispute resolution. Cross-border interoperability is enabled between national governance layers without overriding local property law. This stage addresses gap 11 (post-quantum readiness) at a systemic level because cross-jurisdictional records must remain verifiable under long-term cryptographic migration scenarios.

The governance layer across all five stages is composed primarily of state-authorized institutions: land registries, real-estate regulators, notarial or justice-related authorities, digital identity authorities, and financial regulators where investment or fractional ownership is involved. Banks, inspection bodies, valuation experts, licensed platforms, KYC/AML providers, and regulated custodians participate operationally but do not replace official governance. Their role is to provide verified services within the transaction life-cycle while legal validation remains under the authority of recognized institutions.

9.3 Twelve design requirements

The five-stage pathway defines *when* and *in what sequence* regulated tokenization can evolve. The twelve design require-

ments below define *what* any regulated real-estate tokenization platform must implement at each stage. These requirements are review-derived and design-oriented. They specify minimum conditions for a regulated real-estate tokenization marketplace to become legally meaningful, institutionally acceptable, technically reliable, privacy-preserving, interoperable, secure, and publicly trustworthy.

DR1 — Legal enforceability by design Each token type must carry a clearly defined legal meaning — ownership title, digital certificate, fractional investment claim, lease-related right, financing claim, security-token participation right, or governance right. Token issuance and transfer must be governed by legal instruments, transfer restrictions, approval conditions, and dispute procedures. Without this token-right mapping, a token may move between wallets while the underlying legal right remains unchanged. *Critical from Stage 1; evidence indicators: a legally defined token instrument and transfer-stage legal-effect confirmation.*

DR1 is jurisdiction-invariant: the principle applies regardless of legal tradition, though the specific instrument types (deed, notarial certificate, security prospectus) vary by jurisdiction.

DR2 — Regulator-gated ownership transfer Smart-contract workflows must include approval gates for identity verification, ownership validation, fee and tax confirmation, inspection status, valuation confirmation, payment settlement, and registrar authorization before legal ownership is finalized. Blockchain coordinates; official institutions authorize. *Critical from Stage 2; evidence indicators: smart-contract approval gates and an on-chain registrar authorisation event.*

DR2 is context-sensitive: the specific approval gates and authorising institutions depend on the jurisdiction's registry architecture and regulatory mandate.

DR3 — Identity, KYC/AML, and verifiable authorization The platform must integrate verified digital identity, KYC/AML screening, beneficial-ownership checks, role-based access control, investor suitability checks, and verifiable credentials. Wallet control alone does not establish legal identity, beneficial ownership, or transaction eligibility (Shuaib et al. 2022). Eligibility must be confirmed without exposing unnecessary personal data. *Critical from Stage 2; evidence indicators: wallet-identity linkage and a pre-transfer KYC/AML record.*

DR3 is jurisdiction-invariant: verified identity and KYC/AML compliance are mandatory in any regulated real-estate market, though the specific identity infrastructure varies.

DR4 — Privacy-preserving and crypto-agile certification Sensitive documents — including title deeds, contracts, inspection reports, valuation records, mortgage approvals, identity evi-

dence, and legal correspondence — must remain encrypted in controlled off-chain storage. On-chain anchors may contain hashes, content identifiers, timestamps, or verification proofs. Each anchor must record its algorithm identifier, version, timestamp, issuing authority, document version, and commitment context.

Randomized commitments with controlled opening values should protect predictable documents against candidate-matching attacks while supporting authorized verification. The system must also support selective disclosure, role-based access, correction, revocation, and legally valid re-attestation. A cryptographic transition appends a replacement anchor and an authoritative re-attestation linking the earlier anchor, replacement anchor, and corresponding document version. *Critical from Stage 1; evidence indicators: encrypted off-chain storage, versioned on-chain commitments, controlled opening, and documented correction, revocation, re-anchoring, and re-attestation.*

DR4 is jurisdiction-invariant: privacy-preserving and crypto-agile certification is a minimum condition across jurisdictions, while locally applicable data-protection and evidentiary rules determine its institutional implementation.

DR5 — Fractional ownership and investment-token governance Fractional platforms must define investor rights, governance procedures, revenue-distribution rules, valuation mechanisms, disclosure obligations, dispute procedures, and secondary-transfer conditions before token issuance. Fractional tokenization creates governance questions — voting, maintenance, exit rights, and investor protection — that cannot be resolved by token mechanics alone (Avci and Erzurumlu 2023; Cu et al. 2023; Kranz et al. 2026). *Critical from Stage 3; evidence indicators: pre-issuance investor rights and a documented exit procedure.*

DR5 is context-sensitive: investor-protection rules and governance obligations depend on whether the jurisdiction classifies fractional tokens as securities, collective investment schemes, or digital assets.

DR6 — Conditional workflow and dispute-resolution support Smart contracts must support conditional fund release, refund rules, verified approvals, suspension mechanisms, and escalation procedures. Authorized intervention must be possible when fraud, error, failed inspection, legal dispute, or regulatory violation is detected. Real-estate transactions depend on off-chain events that pure automation cannot resolve. *Critical from Stage 2; evidence indicators: smart-contract suspension and refund logic plus a defined escalation path to an authorized institution.*

DR6 is context-sensitive: the dispute-escalation path and intervening authority depend on whether the jurisdiction operates a common-law, civil-law, or administrative legal system.

DR7 — Legacy-registry and public-sector integration The platform must interoperate with cadastral systems, title registries, valuation databases, digital identity infrastructure, tax systems, inspection systems, financial systems, and legal-document repositories through approved interfaces and auditable workflows. The system must clearly define which record is legally authoritative and how inconsistencies between on-chain evidence and official records are resolved. Blockchain must complement official registries, not compete with them. *Critical from Stage 2; evidence indicators: a documented official-registry interface and an on-/off-chain conflict rule.*

DR7 is context-sensitive: the specific systems to integrate and the legal-effect rules governing them are jurisdiction-specific, though the invariant principle is that blockchain must complement whichever official registry holds legal authority.

DR8 — Scalability, cost control, formal verification, and dynamic threat monitoring Platforms must combine pre-deployment assurance with continuous runtime assurance. Pre-deployment evaluation should include performance benchmarking, transaction-cost analysis, threat modeling, smart-contract auditing, and formal verification for high-risk workflows, including escrow, ownership transfer, investor distribution, and approval logic.

Runtime assurance should monitor on-chain transactions, identity–wallet bindings, authentication and access events, KYC/AML signals, escrow activity, oracle inputs, and authoritative-registry state changes. Deterministic controls can enforce predefined compliance conditions, while AI/ML models can support detection of evolving temporal or graph-based anomalies.

Time-aware analysis of irregular blockchain transaction patterns has supported criminal-address detection in USDT activity on the TRON network (Yadulla et al. 2023). This temporal-detection principle provides a transferable technical basis whose use in regulated real-estate marketplaces requires domain-specific training data, evaluation, and governance. Model outputs should function as explainable risk alerts within auditable review and escalation workflows led by authorized institutional actors.

Evaluation should report throughput, transaction latency, transaction and monitoring cost, precision, recall, F1-score, false-positive rate, detection latency, model drift, adversarial robustness, explanation quality, and response traceability. This requirement is especially important for Stage 4 integrated marketplaces, where transaction volume, transfer restrictions, high asset values, and multi-actor interactions increase operational complexity.

Critical from Stage 3, with continuous monitoring from Stage 4; evidence indicators: performance benchmarks,

audited or formally verified high-risk contracts, documented monitored events and response roles, reported alert precision, recall, false-positive rate and latency, and auditable alert-to-response records.

DR8 remains jurisdiction-invariant at the assurance-principle level: pre-deployment and runtime assurance apply across regulated environments, while monitoring thresholds, authorized roles, and response procedures are configured under the applicable institutional and regulatory workflow.

DR9 — Key management, custody, and recovery Platforms must support regulated custody options, multi-signature authorization, role-based key management, account recovery, emergency freezing, and authorized dispute procedures. Real-estate tokens represent high-value rights; key loss or wallet compromise cannot be treated as a minor technical inconvenience. Custody models must be designed differently for individual buyers, investors, platform operators, regulators, and institutional actors. *Critical from Stage 3; evidence indicators: institutional custody and a tested account-recovery procedure.*

DR9 is jurisdiction-invariant in principle: secure custody and key recovery are necessary for any high-value digital asset, though the licensed custodian models vary by jurisdiction.

DR10 — Interoperability and oracle governance The platform must define approved APIs, data standards, oracle governance rules, cross-platform certification procedures, and regulated bridges. Interoperability must cover registries, identity providers, payment rails, valuation platforms, inspection systems, tax systems, legal databases, and DeFi or cross-chain environments where relevant. *Critical from Stage 4; evidence indicators: documented APIs with at least two external systems and specified oracle-governance rules.*

DR10 is partially context-sensitive: the interoperability principle is invariant, but the specific APIs, data standards, and oracle governance rules must conform to the regulatory standards of the relevant jurisdiction.

DR11 — Long-term cryptographic and evidentiary resilience The platform must maintain an inventory of cryptographic algorithms, keys, signatures, credentials, document commitments, and their dependencies. Signature, identity, authorization, and off-chain key-establishment mechanisms must follow a standards-based migration path toward post-quantum mechanisms, including hybrid operation during transition. Current NIST standards provide the Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM; FIPS 203) and the Module-Lattice-Based Digital Signature Algorithm (ML-DSA; FIPS 204) (National Institute of Standards and Technology 2024b, a).

On-chain commitments must support independent algorithm versioning, periodic security-strength assessment, migration triggers, and append-only re-anchoring. An authorized institution must re-attest the relationship among the earlier commitment, the replacement commitment, and the authoritative document version. Off-chain encryption keys and protected documents must support controlled key rotation and re-encryption where required. These mechanisms preserve the continuity of property evidence while cryptographic standards evolve (Fernández-Caramés and Fraga-Lamas 2020; Barker et al. 2026). *Critical from Stage 4; evidence indicators: a cryptographic inventory, documented algorithms and migration triggers, a tested post-quantum or hybrid transition, and verified sample re-anchoring and institutional re-attestation.*

DR11 is jurisdiction-invariant: long-term cryptographic and evidentiary resilience applies wherever property records must remain verifiable across decades.

DR12 — User-centered marketplace design The marketplace must provide clear transaction-status tracking, plain-language explanations of token meaning, visible approval indicators, risk warnings, user support, and transparent links between blockchain records and official validation. Within the observed sample, the open-response themes and correspondence analysis connect acceptance with understandable services, visible official validation, operational continuity, and rights protection. Prior real-estate blockchain adoption research independently identifies perceived usefulness and ease of use as relevant acceptance conditions (Yeoh et al. 2024; Yang et al. 2025). Users must be able to understand at every step whether they are buying property, investing in a fractional share, verifying a document, paying through escrow, or participating in a regulated marketplace transaction. *Critical from Stage 1; evidence indicators: continuous transaction-status visibility and a plain-language token explanation at the transaction point.*

DR12 is context-sensitive: the specific trust signals, interface conventions, and regulatory endorsement markers that users respond to are shaped by local culture and institutional familiarity, and the survey evidence grounding this requirement reflects the Saudi Arabian respondents surveyed.

9.4 Institutional alignment and framework reach

The six official initiatives discussed in Section 7 provide an institutional applicability assessment of the review-derived framework. Table 15 maps their highest documented functions to the five-stage pathway and design requirements (DRs) and identifies evidence needed for progression (World Bank 2020; Tombs 2019; Lee and Chin 2022; Dubai Land Department 2025; Virtual Assets Regulatory Authority 2026; Saudi Press Agency 2025a, b).

Table 15 Institutional initiatives mapped to framework stages and design requirements

Initiative	Stage coverage evidenced	Coverage status	Requirements reflected	Progression evidence
Sweden	Stage 1 certification and registry anchoring; selected Stage 2 transaction coordination	Stage 1 covered; Stage 2 partially covered	DR3, DR4, DR7; selected DR6	Institution-gated live workflow, dispute procedures, performance evidence, and runtime assurance
Georgia	Stage 1 public-chain anchoring and permissioned registry	Stage 1 covered	DR4, DR7	Identity and KYC integration, workflow governance, interoperability, and operational evaluation
United Kingdom	Stage 1 certification plus Stage 2 identity, payment, smart-contract, and registry coordination	Stage 1 covered; Stage 2 partially covered	DR3, DR6, DR7, DR10, DR12	Live legally effective operation, defined token rights, custody, performance evidence, and runtime monitoring
Singapore	Stage 3 fractional investment, compliance-aware smart contracts, investor verification, and custody	Stage 3 partially covered	DR3, DR5, DR6, DR9, DR10	Authoritative-registry integration, end-to-end governance, dispute procedures, and security evaluation
Dubai	Stage 3 regulated token issuance plus controlled Stage 4 secondary-market functions	Stage 3 covered; Stage 4 partially covered	DR1, DR2, DR3, DR5, DR6, DR12	Integrated national-market operation, registry and service interoperability, custody and recovery, runtime security, and longitudinal performance
Saudi Arabia	Initial Stage 3 title-deed tokenization plus a national architecture directed toward Stage 4	Stage 3 core demonstrated; Stage 4 partially covered in the announced architecture	DR1, DR2, DR3, DR5, DR6, DR7, DR10	Integrated operation, performance evaluation, custody, runtime assurance, cryptographic resilience, and user evaluation

Here, *covered* denotes an explicitly documented core stage function; *partially covered* denotes documented selected functions requiring further evidence across the complete stage and its DRs. Assignments reflect the highest functions supported by the published institutional material.

The mapping distinguishes institutional progression across the framework. Sweden and Georgia centre on Stage 1 certification and registry anchoring, with Sweden also documenting selected Stage 2 coordination. The United Kingdom extends into institution-connected Stage 2 transactions; Singapore demonstrates partial Stage 3 regulated fractional investment; Dubai combines covered Stage 3 issuance with partial Stage 4 secondary-market functions; and Saudi Arabia combines initial Stage 3 title-deed tokenization with an announced architecture partially covering Stage 4. Thus, all six initiatives fall within Stages 1–4, while complete Stage 4 and Stage 5 readiness remain progression thresholds. Complete Stage 4 requires end-to-end national marketplace integration, full registry and service interoperability, custody and recovery, dispute handling, continuous runtime assurance, and longitudinal performance. Stage 5 further requires mutual legal recognition, interoperable national registries and identity systems, compliant cross-border settlement, coordinated investor protection, cross-jurisdictional dispute resolution, runtime security assurance, and long-term cryptographic resilience.

Current initiatives provide substantial evidence of registry integration, identity verification, regulated tokenization, institutional coordination, and controlled market access. Their documentation provides less developed evidence of end-to-end performance benchmarking, formal verification, continuous threat monitoring, custody recovery, long-term cryptographic resilience, and cross-jurisdictional interoperability. The framework therefore supplies a common progression model, identifies fully or partially reflected stages and requirements, and specifies the technical and governance evidence needed for a secure, sustainable regulated marketplace.

9.5 RQ7: The socio-technical governance framework as a design contribution

The synthesis of 63 primary studies, 153 survey responses, twelve research gaps, and the cross-initiative applicability assessment yields a *jurisdiction-gated governance framework* for regulated real-estate tokenization, in which blockchain functions as institutional trust-support infrastructure within recognized legal authority. The framework organizes current technical and institutional advances into a five-stage progression from trusted digital certification to federated cross-jurisdictional operation.

The framework makes four bounded contributions. First, it treats blockchain-based real-estate tokenization as a regulated system rather than as an isolated blockchain application, integrating architectural, legal, institutional, market, and adoption dimensions while distinguishing jurisdiction-invariant from context-sensitive requirements. Second, it makes the derivation of the design requirements traceable: each requirement is linked to systematic-review evidence, a documented deployment gap, and, where applicable, an exploratory survey-informed implication. Third, the five-stage jurisdiction-gated pathway provides sequencing logic that maps progression to gap resolution, allowing incremental adoption without bypassing legal or institutional requirements. Fourth, the framework specifies review-derived minimum conditions for a regulated real-estate-tokenization marketplace to become legally meaningful, institutionally accountable, technically reliable, privacy-preserving, interoperable, continuously monitored, cryptographically resilient, and publicly trustworthy. These contributions are design-oriented propositions whose empirical validation proceeds through future expert, case-study, and implementation evaluation.

Blockchain in this model is not the source of legal authority. It is the infrastructure through which legal authority, market coordination, and public trust become more transparent, auditable, and operationally efficient. Real-estate tokenization succeeds not when it replaces institutions, but when it makes institutions more capable of governing digital property markets.

10 Study limitations and threats to validity

The scope and interpretation of this study are bounded by eight validity considerations covering evidence retrieval, coding, evidence maturity, survey design, perception measurement, qualitative-response depth, jurisdictional transferability, and technology and regulatory evolution, addressed through established systematic-review and research-quality practices (Kitchenham et al. 2009; Page et al. 2021).

T1 — Search and database coverage Indexing limits, varied terminology, and publication delays can affect retrieval across blockchain real estate, land registries, property or RWA tokenization, NFTs, security tokens, and digital-asset marketplaces. Major databases, citation chaining, Elicit, recent-preprint scanning, and manual screening provided complementary coverage.

T2 — Design artifact construction and coding validity Studies often span registration, NFTs, smart contracts, escrow,

onboarding, certification, and marketplace functions, making dominant architectural assignment interpretive. The eight-dimensional taxonomy standardized coding, with the joint-consensus procedure reported in Section 4.4.

T3 — evidence maturity The corpus primarily comprises concepts, architectures, simulations, and prototypes, so findings represent academic and prototype-oriented evidence. Advanced fractional, security-token, DeFi, and hybrid-market uses still require legal validation, institutional integration, security testing, interoperability, and user acceptance. Reported quality and maturity distributions preserve this distinction.

Official self-reported disclosures complement the corpus by documenting the scope and status of supervised initiatives. Independent evaluation remains the basis for confirming performance, security, sustained operation, and complete stage attainment; framework coverage is therefore assigned conservatively to explicitly documented functions.

T4 — Survey representativeness The 153 valid responses were recruited through social media and targeted contact lists, establishing exploratory, sample-specific interpretation. Recorded variables focused on aggregate perceptions, reserving age, education, blockchain familiarity, real-estate experience, professional role, stratified recruitment, and subgroup analysis for external-validity testing in subsequent studies.

T5 — Shared method and perception–behavior relationship Cronbach's alpha (0.933) and the first-component share (68.7%) indicate a strong general response orientation within the single self-report instrument. The differentiated-profile sensitivity assessment retained high consistency ($\alpha = 0.884$), a 55.8% first-component share, and positive Spearman coefficients. Pairwise associations are therefore used as exploratory evidence of relative orientation, while the open–closed correspondence supplies item-specific interpretation. Confirmatory work can use validated multi-item constructs, stratified samples, longitudinal designs, and observed behavior.

T6 — Open-ended response depth Short open-ended responses supported thematic interpretation of advantages and concerns. Structured interviews or focus groups with buyers, sellers, investors, regulators, brokers, banks, and operators can provide deeper qualitative evidence.

T7 — Generalizability across legal contexts Real-estate law, registration, notarial systems, and public governance vary across jurisdictions. Saudi institutional features shape the survey patterns and context-sensitive requirements. The framework therefore classifies **DR1**, **DR3**, **DR4**, **DR8**, **DR9**,

and **DR11** as jurisdiction-invariant, while **DR2**, **DR5**, **DR6**, **DR7**, **DR10**, and **DR12** require adaptation to local registries, legal traditions, and mandates. Section 11 prioritizes multi-jurisdictional validation.

T8 — Technology and regulatory evolution Platforms, token and identity standards, privacy, custody, interoperability, AI/ML security monitoring, post-quantum cryptography, and legal treatment continue to evolve. Technology-neutral principles cover enforceability, registries, identity, privacy, governance, custody, interoperability, formal verification, continuous monitoring, governed response, and crypto-agility. Domain evaluation, model-drift monitoring, and periodic control reassessment support continued relevance.

These considerations bound the contribution as systematic, exploratory, and design-oriented. A staged pilot can compare the framework with an existing workflow under equivalent transaction, registry, compliance, security, and service conditions, combining technical benchmarks, lifecycle cost, stakeholder adoption, process outcomes, and institutional assessment.

11 Conclusions

This study combined a PRISMA-based review of 63 primary studies with exploratory perceptions from 153 respondents in a highly regulated market. The evidence traces progression from registry-centered certification to fractional investment and integrated marketplaces. Peer-reviewed work remains prototype- and simulation-dominant, while official initiatives partially progress through Stage 4; complete Stage 4 operation and Stage 5 readiness require independent operational validation. Respondents expressed a coherent, predominantly cautious orientation. Reliability and single-factor findings locate the positive associations within a general response disposition, while the open–closed analysis connects recognition of particular capabilities with conditional adoption. Its emphasis on legal authority, operational assurance, and user protection strengthens the staged framework. Regulated tokenization is therefore a socio-technical challenge aligning technical mechanisms, recognized rights, authoritative registries, institutional accountability, and public trust.

The contribution links an eight-dimensional taxonomy, twelve deployment gaps, regulated-market perceptions, and a jurisdiction-gated framework. Its five stages and twelve requirements address token rights, enforceability, identity/KYC/AML, privacy, valuation, governance, disputes, scalability and lifecycle cost, custody, interoperability, dynamic monitoring, cryptographic resilience, and user-centered design. They structure progression from certification through

regulated transactions and tokenization to national and cross-jurisdictional marketplaces.

Policy makers, regulators, and land authorities should define token rights before issuance and retain authoritative registries. Fractional or security-token models further require legal classification, valuation, investor protection, custody, disputes, and registry integration. Operators and financial institutions should provide identity and KYC/AML, privacy-preserving documents, recovery, smart-contract assurance, continuous monitoring, interoperability, and measurable lifecycle-cost and service targets. Staged pilots can test these conditions under regulatory and operational control.

Future expert assessments, cross-jurisdictional cases, and operational pilots should compare performance, availability, lifecycle cost, integration and compliance effort, process outcomes, and adoption with existing workflows. Priorities also include domain-specific anomaly detection, formal verification, privacy-preserving and crypto-agile document anchoring, and post-quantum migration and re-attestation. Blockchain creates its greatest real-estate value by strengthening the institutions that establish, transfer, protect, and govern property rights.

Appendix A Review coding, quality assessment, and evidence-maturity protocol

Tables 16–19 summarize the data-extraction fields, taxonomy coding scheme, quality-assessment rubric, and evidence-maturity classification, respectively. This appendix documents the protocol applied to the final 63-study corpus: data-extraction fields, taxonomy coding, quality assessment, and evidence maturity. It complements the concise methodology in Section 4 and follows the taxonomy in Section 3 and systematic-review guidance (Kitchenham et al. 2009; Petersen et al. 2015). The protocol generated a 63-row extraction matrix, a 315-judgment quality matrix, and a maturity classification, supporting the results in Section 5, evidence-strength interpretation, and the cross-dimensional synthesis in Section 7.

A.1 Data-extraction fields

The extraction matrix contained 20 fields grouped into three categories, producing a maximum of $63 \times 20 = 1,260$ coded entries. Fields not explicitly reported in a study were coded as *not reported* rather than inferred. When a study addressed more than one category, the dominant category was recorded as the primary classification and secondary categories retained in coding notes.

Table 16 Data-extraction fields applied to the 63-study corpus. Bold indicates final/aggregate total values

Field group	Extraction fields	Coding purpose
Descriptive	Publication year; venue type; study type or research method; application or property focus; evaluation method; reported limitations.	Bibliographic and methodological profile to support publication-trend and evidence-profile analysis.
Taxonomy	Architectural family; ownership model; transaction type; tokenization model; certification mechanism; stakeholder structure; technology model; adoption barrier.	Classify each study per the taxonomy in Section 3 and support systematic review results.
Interpretive	Evidence maturity; regulatory treatment; identity/KYC treatment; privacy treatment; security assumptions; long-term security readiness.	Support interpretation of maturity, deployability, compliance, privacy, identity, and long-term risk.
Total	20 extraction fields	Up to 1,260 coded entries across 63 studies.

A.2 Taxonomy coding scheme

The architectural family was coded as a single dominant category per study. Other dimensions could include secondary notes when a study addressed multiple aspects.

Studies scoring 4.0–5.0 were treated as higher-quality evidence, 2.5–3.5 as moderate-quality, and below 2.5 as lower-quality. No included study fell below 2.5; the rubric specified that any lower-quality study satisfying the inclusion criteria would be interpreted cautiously during synthesis.

Table 17 Taxonomy coding scheme used for the included studies

Taxonomy dimension	Coding categories
Architectural family	Registry-centric systems; NFT-based ownership systems; fractional ownership and tokenized investment systems; rental/leasing/mortgage automation systems; hybrid regulated marketplace architectures; DeFi-oriented investment-token frameworks.
Ownership model	Full ownership or title record; transactional property rights; NFT-based unique ownership; fractional ownership or shared investment; rental or lease rights; financing-related claims; security-token participation rights.
Transaction type	Registration; buying or selling; ownership transfer; registry update; renting or leasing; mortgage or collateral; financing or investment; resale or secondary trading; settlement; investor onboarding; dispute resolution.
Tokenization model	No token; NFT; fungible or fractional token; hybrid token model; security-token-oriented model; ERC-20/EIP-1400-oriented model.
Certification mechanism	On-chain record or hash; IPFS or decentralized storage; content identifier; oracle-supported verification; digital signatures or SSI credentials; authoritative registry or government verification; regulator-approved certification; prospectus/SPV validation.
Stakeholder structure	Buyer or seller; investor, tenant, or landlord; registrar, land authority, or notary; regulator; financial institution; inspection or valuation body; platform operator; issuer, developer, or SPV; KYC/AML provider; validator, technology provider, or custody provider.
Technology model	Public or permissioned blockchain; Ethereum or Polygon; Hyperledger Fabric; smart contracts; escrow contracts; IPFS or off-chain storage; oracles; privacy or ZKP mechanisms; DeFi integration; PoA, GPoA, or PBFT consensus; SSI, DIDs, or verifiable credentials; cross-chain or multi-chain integration; post-quantum-ready mechanisms.
Adoption barrier	Legal enforceability or regulatory classification; KYC/AML and identity binding; privacy; valuation; governance; dispute resolution; registry or legacy-system integration; scalability and cost; verification, cybersecurity, or key management; interoperability; stakeholder acceptance or public trust; custody and investor protection.

A.3 Quality-assessment rubric

Each criterion was scored 0 (not addressed), 0.5 (partially addressed), or 1 (adequately addressed), producing a maximum of 5 per study and $63 \times 5 = 315$ criterion-level judgments across the corpus. Scores interpreted evidence strength rather than excluding studies after inclusion.

A.4 Evidence-maturity classification

Each study was assigned one primary maturity level based on its strongest form of evidence. Where a study provided multiple evidence types, it was coded at the highest level present.

Table 18 Quality-assessment rubric. Bold indicates final/aggregate total values

Code	Quality criterion	Score
QA1	Clarity of the research objective.	0–1
QA2	Relevance to blockchain-based real-estate tokenization.	0–1
QA3	Clarity of the proposed architecture, framework, model, or method.	0–1
QA4	Adequacy of evaluation or supporting evidence.	0–1
QA5	Explicit discussion of limitations, risks, or assumptions.	0–1
Total	Maximum quality score per study.	5

Table 19 Evidence-maturity levels applied to the 63-study corpus

Maturity level	Coding rule
Conceptual proposal	Presents a concept, argument, or high-level model without detailed architecture, implementation, or evaluation.
Architecture or framework model	Provides a structured architecture, process model, design framework, or system specification without full implementation.
Prototype or proof of concept	Implements a partial or complete prototype, smart contract, application workflow, or demonstrator.
Simulation or experimental evaluation	Evaluates cost, performance, gas usage, scalability, or feasibility through simulation or experiments.
Case study or scenario evaluation	Applies the model to a case, jurisdiction, real-estate process, scenario, or platform context.
Empirical or expert-based evaluation	Uses survey data, interviews, expert validation, market data, or empirical analysis.
Deployed or operational system	Reports an operational platform or deployed system with evidence beyond prototype demonstration.

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Data Availability The structured 63-study extraction matrix, quality-appraisal matrix, and masked survey answers supporting this study are publicly available in the GitHub repository TokensReview at <https://github.com/humaimedy/TokensReview>. The masked survey workbook contains the eight Likert-item answers and two open-ended answers for the 153-response dataset used in the reported quantitative, sensitivity, and open-closed correspondence analyses.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics approval and consent to participate This study involved an anonymous, minimal-risk perception survey of adult participants aged 18 years or older. Participation was voluntary, no personally identifiable information was collected, and respondents provided informed consent before submitting the questionnaire.

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